1. Personal Portfolio Website using HTML and CSS

Program Coding:

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Portfolio - Home</title>
  <style>
    body {
      margin: 0;
      font-family: 'Segoe UI', sans-serif;
      background-color: #f1f5f9;
    }
    /* Navigation Bar */
    nav {
      background-color: #0f172a;
      color: white;
      display: flex;
      justify-content: space-between;
      align-items: center;
      padding: 15px 60px;
    }
    nav h2 {
      margin: 0;
      font-size: 22px;
      letter-spacing: 1px;
    }
    nav ul {
      list-style: none;
      display: flex;
      gap: 25px;
      margin: 0;
      padding: 0;
    }
    nav ul li a {
      text-decoration: none;
      color: #cbd5e1;
      font-weight: 500;
      transition: 0.3s;
    }
    nav ul li a:hover {
      color: #2563eb;
    }
    /* Hero Section */
    .home {
      height: 90vh;
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      text-align: center;
      background: linear-gradient(to right,
#1e293b, #0f172a);
      color: white;
    }
    .home h1 {
      font-size: 48px;
      margin-bottom: 10px;
    }
    .home p {
      font-size: 20px;
      color: #cbd5e1;
      margin-bottom: 30px;
    }
    .home button {
      padding: 12px 30px;
      font-size: 16px;
      background-color: #2563eb;
      color: white;
      border: none;
      border-radius: 6px;
      cursor: pointer;
      transition: 0.3s;
    }
    .home button:hover {
      background-color: #1d4ed8;
      transform: scale(1.05);
```

```
    }
  </style>
</head>
<body>
<nav>
  <h2>My Portfolio</h2>
  <ul>
    <li><a
href="index.html">Home</a></li>
    <li><a
href="professional.html">Professional
Info</a></li>
    <li><a
href="achievements.html">Achievements</
a></li>
    <li><a
href="experience.html">Experience</a></li
>
  </ul>
</nav>
<section class="home">
  <h1>Hello, I'm Varshini selvakumar</h1>
  <p>Computer Science Student |
Front-End Developer</p>
  <button>Contact Me</button>
</section>
</body>
</html>
```

Professional.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Professional Information</title>
  <style>
    body {
      margin: 0;
      font-family: 'Segoe UI', sans-serif;
      background-color: #f1f5f9;
    }
    /* Navigation Bar */
    nav {
      background-color: #0f172a;
      color: white;
      display: flex;
      justify-content: space-between;
      align-items: center;
      padding: 15px 60px;
    }
    nav h2 {
      margin: 0;
      font-size: 22px;
      letter-spacing: 1px;
    }
    nav ul {
      list-style: none;
      display: flex;
      gap: 25px;
      margin: 0;
      padding: 0;
    }
    nav ul li a {
      text-decoration: none;
      color: #cbd5e1;
      font-weight: 500;
      transition: 0.3s;
    }
    nav ul li a:hover {
      color: #2563eb;
    }
    /* Page Title */
    .page-title {
      text-align: center;
      padding: 40px 20px 20px 20px;
      color: #0f172a;
    }
    .page-title h1 {
      font-size: 36px;
      margin-bottom: 10px;
    }
    /* Content Section */
    .container {
```

```
            width: 80%;
            margin: auto;
            padding: 20px 0 60px 0;
            display: grid;
            grid-template-columns:
repeat(auto-fit, minmax(300px, 1fr));
            gap: 25px;
        }
        .card {
            background-color: white;
            padding: 25px;
            border-radius: 10px;
            box-shadow: 0 5px 15px
rgba(0,0,0,0.1);
            transition: 0.3s;
        }
        .card:hover {
            transform: translateY(-5px);
        }
        .card h3 {
            margin-top: 0;
            color: #2563eb;
        }
        .card p, .card li {
            color: #334155;
            line-height: 1.6;
        }
        ul {
            padding-left: 20px;
        }
    </style>
</head>
<body>
<nav>
    <h2>My Portfolio</h2>
    <ul>
        <li><a
href="index.html">Home</a></li>
        <li><a
href="professional.html">Professional
Info</a></li>
        <li><a
href="achievements.html">Achievements</
a></li>
        <li><a
href="experience.html">Experience</a></li
>
    </ul>
</nav>
<div class="page-title">
    <h1>Professional Information</h1>
</div>
<div class="container">
    <div class="card">
        <h3>🎓 Education</h3>
        <p><strong>Bachelor in Computer
Science</strong></p>
        <p>PSGR Krishnammal college for
women</p>
        <p>2023 - 2026</p>
    </div>
    <div class="card">
        <h3>💻 Technical Skills</h3>
        <ul>
            <li>HTML & CSS</li>
            <li>JavaScript</li>
            <li>Python</li>
            <li>C++</li>
        </ul>
    </div>
    <div class="card">
        <h3>🎯 Career Objective</h3>
        <p>
            Motivated Computer Science student
seeking opportunities
            to apply technical skills in web
development and contribute
            to innovative projects.
        </p>
    </div>
</div>
</body> </html>
```

Achievements.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Achievements</title>
  <style>
    body {
      margin: 0;
      font-family: 'Segoe UI', sans-serif;
      background-color: #f1f5f9;
    }
    /* Navigation Bar */
    nav {
      background-color: #0f172a;
      color: white;
      display: flex;
      justify-content: space-between;
      align-items: center;
      padding: 15px 60px;
    }
    nav h2 {
      margin: 0;
      font-size: 22px;
      letter-spacing: 1px;
    }
    nav ul {
      list-style: none;
      display: flex;
      gap: 25px;
      margin: 0;
      padding: 0;
    }
    nav ul li a {
      text-decoration: none;
      color: #cbd5e1;
      font-weight: 500;
      transition: 0.3s;
    }
    nav ul li a:hover {
      color: #2563eb;
    }
    /* Page Title */
    .page-title {
      text-align: center;
      padding: 40px 20px 20px 20px;
      color: #0f172a;
    }
    .page-title h1 {
      font-size: 36px;
      margin-bottom: 10px;
    }
    /* Achievement Cards */
    .container {
      width: 80%;
      margin: auto;
      padding: 20px 0 60px 0;
      display: grid;
      grid-template-columns:
repeat(auto-fit, minmax(300px, 1fr));
      gap: 25px;
    }
    .card {
      background-color: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 5px 15px
rgba(0,0,0,0.1);
      transition: 0.3s;
    }
    .card:hover {
      transform: translateY(-5px);
    }
    .card h3 {
      margin-top: 0;
      color: #2563eb;
    }
    .card p {
      color: #334155;
      line-height: 1.6;
    }
    .highlight {
      font-weight: bold;
```

```
        color: #0f172a;
      }
    </style>
</head>
<body>
<nav>
    <h2>My Portfolio</h2>
    <ul>
      <li><a
href="index.html">Home</a></li>
      <li><a
href="professional.html">Professional
Info</a></li>
      <li><a
href="achievements.html">Achievements</
a></li>
      <li><a
href="experience.html">Experience</a></li
>
    </ul>
</nav>
<div class="page-title">
    <h1>My Achievements</h1>
</div>
<div class="container">
    <div class="card">
      <h3>🏆 Coding Competition
Winner</h3>
      <p>
        Secured <span class="highlight">1st
Prize</span> in Inter-College Coding
Competition 2024.
      </p>
    </div>
    <div class="card">
      <h3>📜 Web Development
Certification</h3>
      <p>
        Successfully completed a certified
course in Full Stack Web Development.
      </p>
    </div>
    <div class="card">
      <h3>🎯 Academic Excellence</h3>
      <p>
        Achieved <span
class="highlight">95% marks</span> in
Final Year Project.
      </p>
    </div>
    <div class="card">
      <h3>🚀 Hackathon Participant</h3>
      <p>
        Participated in National Level
Hackathon and developed an innovative web
solution.
      </p>
    </div>
</div>
</body>
</html>
```

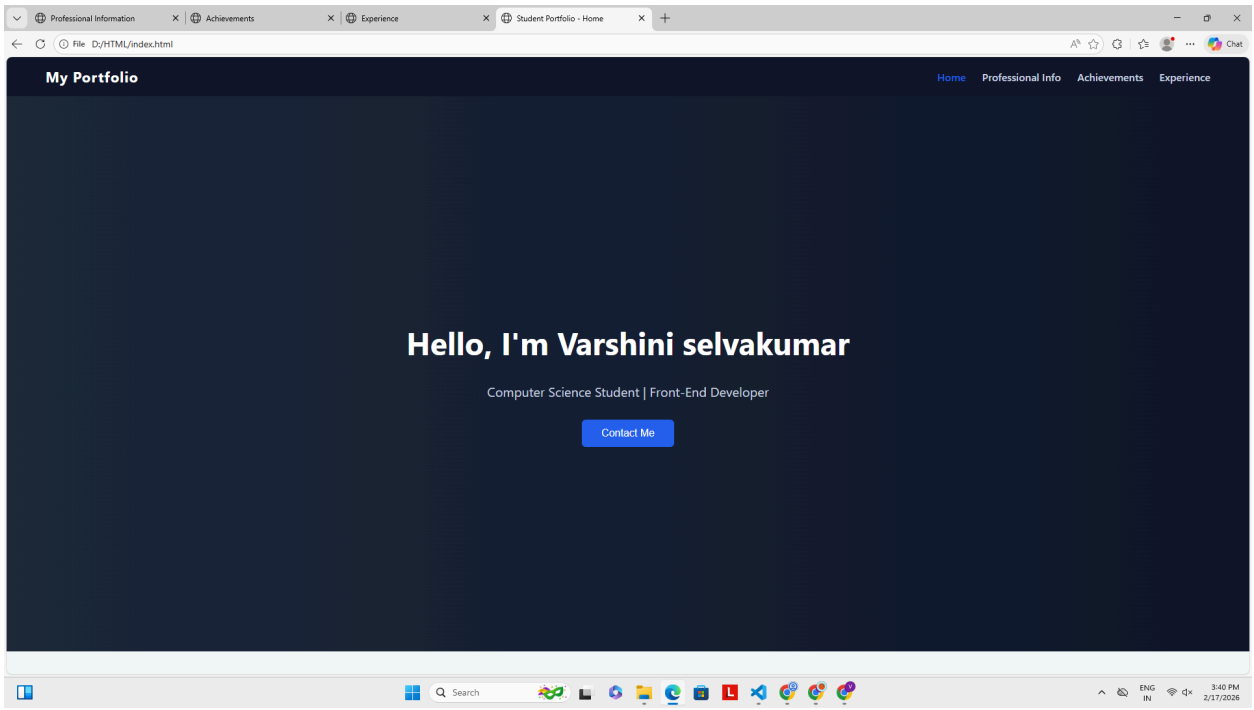
Experience.html:

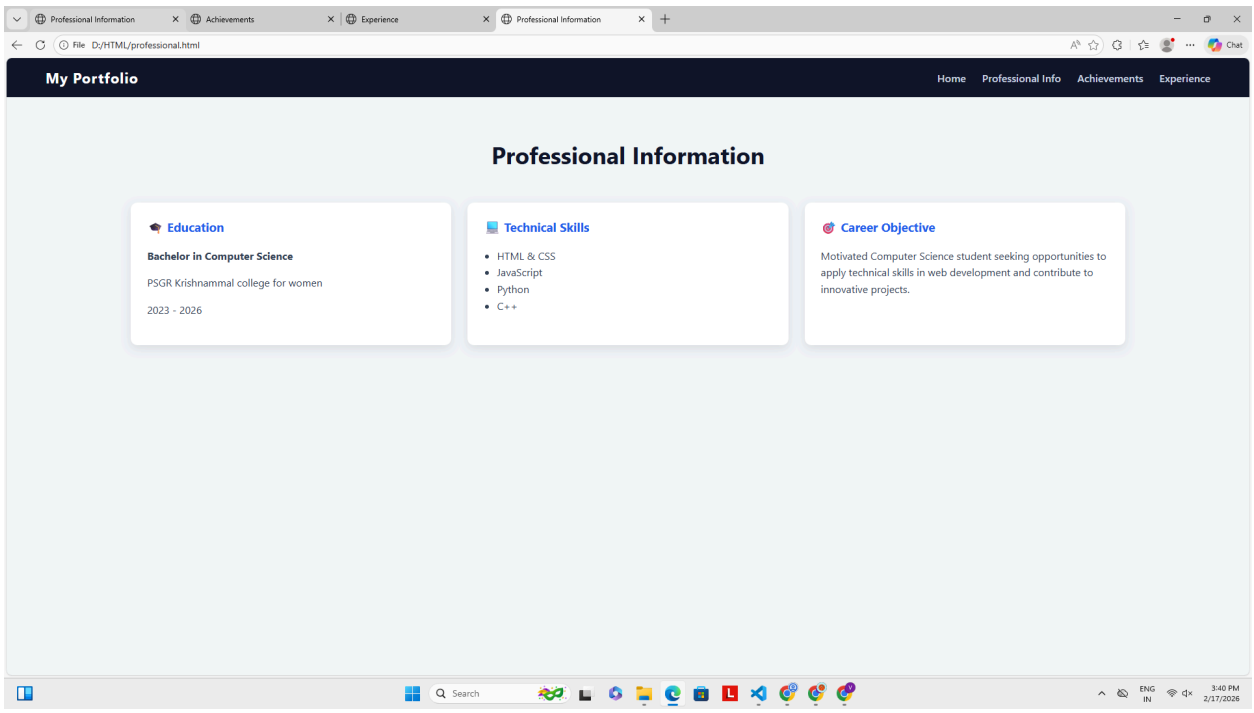
```
<!DOCTYPE html>
<html>
<head>
    <title>Experience</title>
    <style>
      body {
        margin: 0;
        font-family: 'Segoe UI', sans-serif;
        background-color: #f1f5f9;
      }
      /* Navigation Bar */
      nav {
        background-color: #0f172a;
        color: white;
        display: flex;
        justify-content: space-between;
        align-items: center;
        padding: 15px 60px;
      }
```

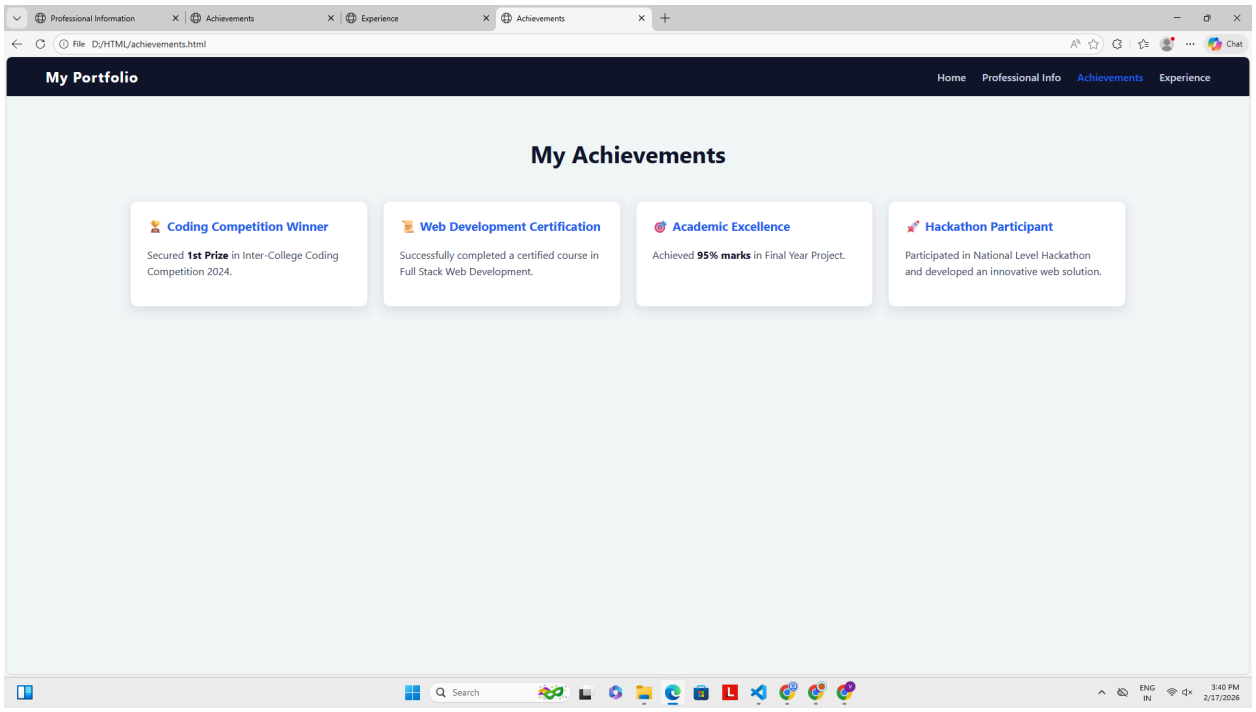
```
nav h2 {
    margin: 0;
    font-size: 22px;
    letter-spacing: 1px;
}
nav ul {
    list-style: none;
    display: flex;
    gap: 25px;
    margin: 0;
    padding: 0;
}
nav ul li a {
    text-decoration: none;
    color: #cbd5e1;
    font-weight: 500;
    transition: 0.3s;
}
/*nav ul li a:hover {
    color: #2563eb;
}*/
/* Page Title */
.page-title {
    text-align: center;
    padding: 40px 20px 20px 20px;
    color: #0f172a;
}
.page-title h1 {
    font-size: 36px;
    margin-bottom: 10px;
}
/* Timeline Section */
.timeline {
    width: 80%;
    margin: 40px auto 80px auto;
    position: relative;
}
.timeline::after {
    content: '';
    position: absolute;
    width: 4px;
    /*background-color: #0a0a0a;*/
    top: 0;
    bottom: 0;
    left: 50%;
    margin-left: -2px;
}
.container {
    padding: 10px 20px;
    position: relative;
    background-color: inherit;
    width: 50%;
}
.container.left {
    left: 0;
}
.container.right {
    left: 50%;
}
.content {
    padding: 10px;
    background-color: white;
    border-radius: 4px;
    box-shadow: 0 5px 15px
rgba(0,0,0,0.1);
    transition: 0.3s;
}
.content:hover {
    transform: translateY(-5px);
}
.content h3 {
    margin-top: 0;
    color: #2563eb;
}
.content p {
    color: #334155;
    line-height: 1.6;
}
/* Responsive */
@media screen and (max-width:
768px) {
    .timeline::after {
```

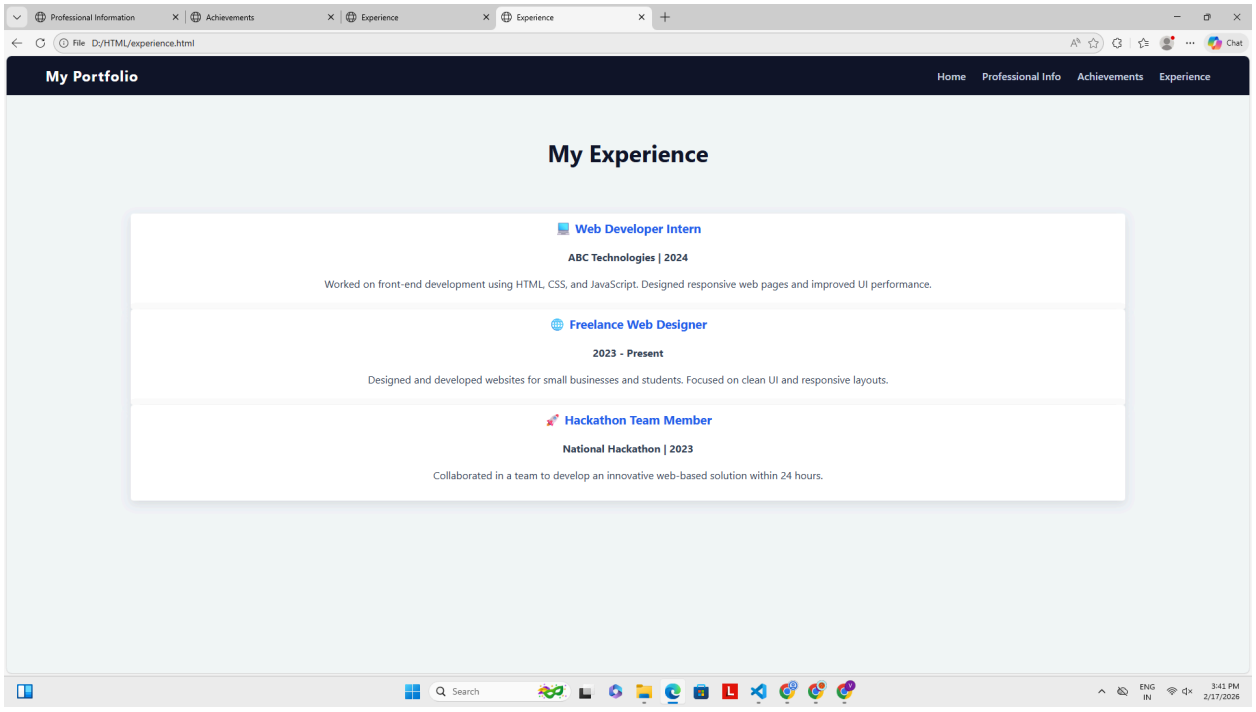
```
            left: 31px;
        }
        .container {
            width: 75%;
            padding-left: 50px;
            padding-right: 10px;
        }
        .container.right {
            left: 0%;
        }
    }
  </style>
</head>
<body>
<nav>
  <h2>My Portfolio</h2>
  <ul>
    <li><a
href="index.html">Home</a></li>
    <li><a
href="professional.html">Professional
Info</a></li>
    <li><a
href="achievements.html">Achievements</
a></li>
    <li><a
href="experience.html">Experience</a></li
>
  </ul>
</nav>
<div class="page-title">
  <h1>My Experience</h1>
</div>
<div class="timeline">
  <center>
    <div class="content">
      <h3>💻 Web Developer Intern</h3>
      <p><strong>ABC Technologies |
2024</strong></p>
      <p>Worked on front-end
development using HTML, CSS, and
JavaScript. Designed responsive web pages
and improved UI performance.</p>
    </div>
      <div class="content">
      <h3>🌐 Freelance Web
Designer</h3>
      <p><strong>2023 -
Present</strong></p>
      <p>Designed and developed
websites for small businesses and students.
Focused on clean UI and responsive
layouts.</p>
    </div>
     <div class="content">
      <h3>🚀 Hackathon Team
Member</h3>
      <p><strong>National Hackathon |
2023</strong></p>
      <p>Collaborated in a team to
develop an innovative web-based solution
within 24 hours.</p>
    </div>
  </center>
</div>
</body>
</html>
```

Output:


Professional Information
Achievements
Experience
Student Portfolio - Home
File D:/HTML/index.html
Chat
My Portfolio
Home
Professional Info
Achievements
Experience
Hello, I'm Varshini selvakumar
Computer Science Student | Front-End Developer
Contact Me
Search
ENG
IN
3:40 PM
2/17/2026


Professional Information
Achievements
Experience
Professional Information
File D:/HTML/professional.html
Chat
My Portfolio
Home
Professional Info
Achievements
Experience
Professional Information
Education
Bachelor in Computer Science
PSGR Krishnammal college for women
2023 - 2026
Technical Skills
HTML & CSS
JavaScript
Python
C++
Career Objective
Motivated Computer Science student seeking opportunities to apply technical skills in web development and contribute to innovative projects.
Search
ENG
IN
3:40 PM
2/17/2026


Professional Information
Achievements
Experience
Achievements
File D:/HTML/achievements.html
Chat
My Portfolio
Home
Professional Info
Achievements
Experience
My Achievements
Coding Competition Winner
Secured 1st Prize in Inter-College Coding Competition 2024.
Web Development Certification
Successfully completed a certified course in Full Stack Web Development.
Academic Excellence
Achieved 95% marks in Final Year Project.
Hackathon Participant
Participated in National Level Hackathon and developed an innovative web solution.
Search
ENG
IN
3:40 PM
2/17/2026


Professional Information
Achievements
Experience
Experience
File D:/HTML/experience.html
Chat
My Portfolio
Home
Professional Info
Achievements
Experience
My Experience
Web Developer Intern
ABC Technologies | 2024
Worked on front-end development using HTML, CSS, and JavaScript. Designed responsive web pages and improved UI performance.
Freelance Web Designer
2023 - Present
Designed and developed websites for small businesses and students. Focused on clean UI and responsive layouts.
Hackathon Team Member
National Hackathon | 2023
Collaborated in a team to develop an innovative web-based solution within 24 hours.
Search
ENG
IN
3:41 PM
2/17/2026

2. Smart Shopping Cart Application using React.js Components

Program Coding:

App.jsx:

```
import { useState } from "react"; //State
import ProductCard from
"./ProductCard.jsx";
import Cart from "./Cart.jsx";
import "./App.css";

// Functional Components
function App() {
  const products = [
    { id: 1, name: "Headphones", price: 1500
},
    { id: 2, name: "Smart Watch", price: 2500
},
    { id: 3, name: "Bluetooth Speaker", price:
3000 }
  ];

  const [cart, setCart] = useState([]); //State

  const addToCart = (product) => {
    setCart([...cart, product]);
  };

  const removeFromCart =
(indexToRemove) => {
    setCart(cart.filter((_, index) => index !==
indexToRemove));
  };

  return (
    <div className="container">
      <h1>🛒 Smart Shopping Cart</h1>

      <div className="layout">
        <div>
          <h2>Products</h2>
          <div className="product-list">
            {products.map((product) => (
              //Props
              <ProductCard
                key={product.id}
                product={product}
                addToCart={addToCart}
              />
            ))}
          </div>
        </div>
        <Cart cart={cart}
removeFromCart={removeFromCart} />
//Props
      </div>
    </div>
  );
}

export default App;
```

Index.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./index.css";

const root =
ReactDOM.createRoot(document.getElemen
tById("root"));
root.render(<App />);
```

ProductCard.jsx:

```
//Functional Components
function ProductCard({ product, addToCart }) {
  return (
    <div className="product-card">
      <h3>{product.name}</h3>
      <p
className="price">₹{product.price}</p>
      <button onClick={() =>
addToCart(product)}>
        Add to Cart
      </button>
    </div>
  );
}

export default ProductCard;
```

Cart.jsx:

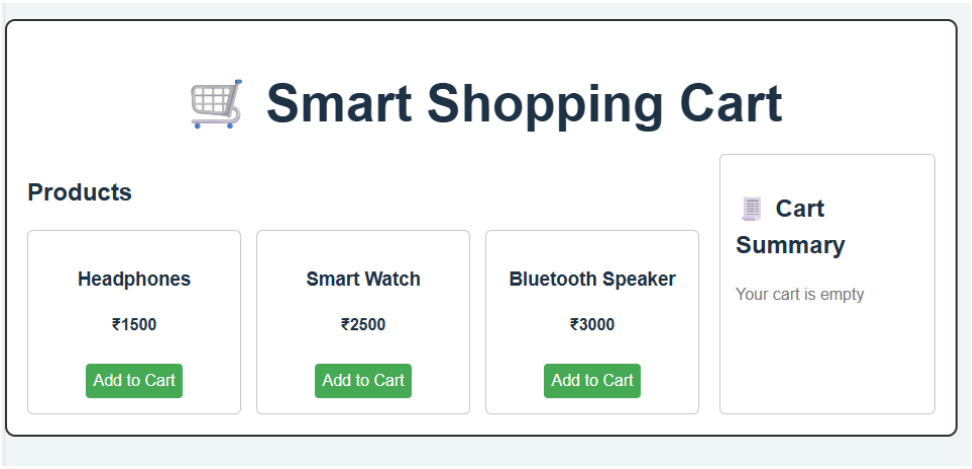
```
//Functional Components
function Cart({ cart, removeFromCart }) {
  const total = cart.reduce((sum, item) =>
sum + item.price, 0);

  return (
    <div className="cart">
      <h2>🧾 Cart Summary</h2>

      {cart.length === 0 ? (
        <p className="empty">Your cart is
empty</p>
      ) : (
        <>
          <ul>
            {cart.map((item, index) => (
              <li key={index}>
                <span>
                  {item.name} – ₹{item.price}
                </span>
                <button
                  className="remove"
                  onClick={() =>
removeFromCart(index)}
                >
                  Remove
                </button>
              </li>
            ))}
          </ul>

          <h3>Total: ₹{total}</h3>
        </>
      )}
    </div>
  );
}

export default Cart;
```

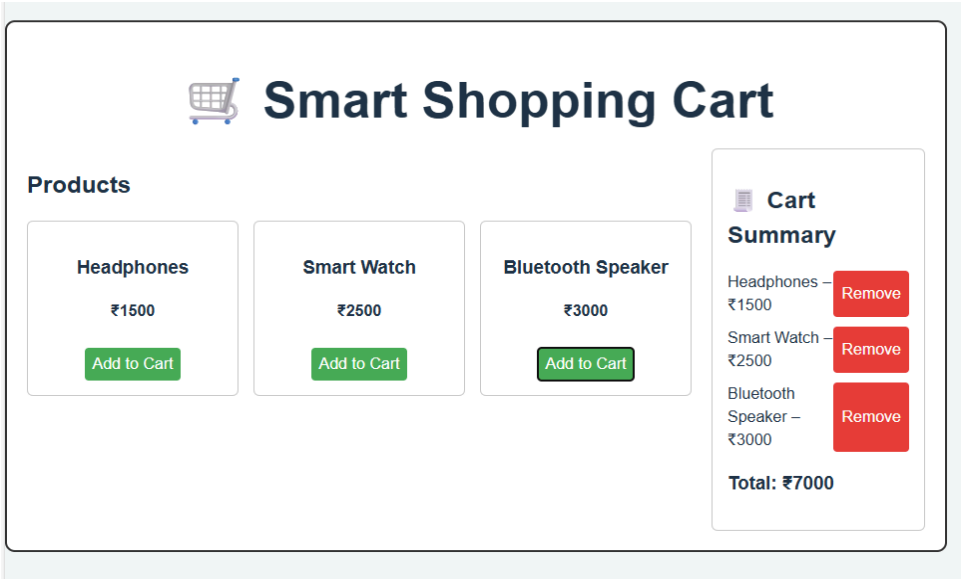
App.css:

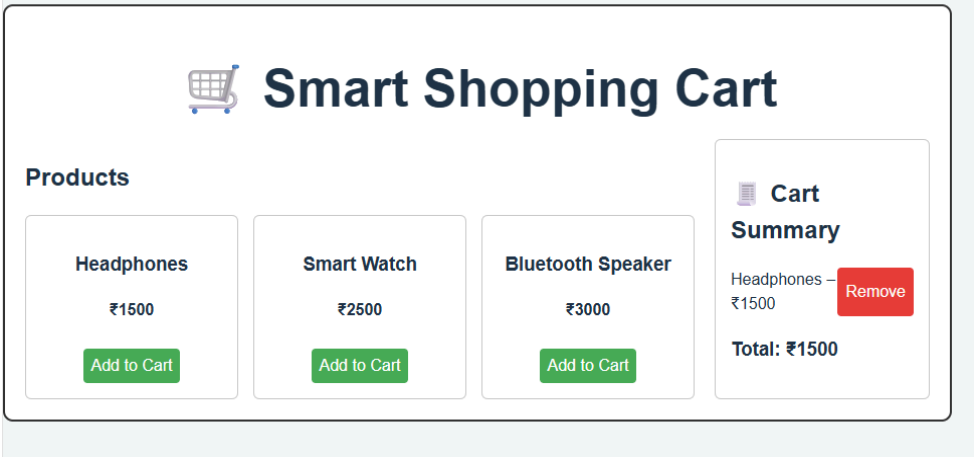
```
body {
  margin: 0;
  background-color: #f4f6f8;
}

/* OUTER BORDER */
.container {
  font-family: Arial, sans-serif;
  width: 900px;
  margin: 40px auto;
  padding: 20px;
  background-color: white;
  border: 2px solid #333;
  border-radius: 10px;
}

h1 {
```

```
  text-align: center;
  margin-bottom: 20px;
}

.layout {
  display: flex;
  gap: 20px;
}

/* PRODUCTS */
.product-list {
  display: flex;
  gap: 15px;
}

.product-card {
  border: 1px solid #ccc;
  border-radius: 6px;
  padding: 15px;
  width: 180px;
  text-align: center;
}

.product-card .price {
  font-weight: bold;
}

.product-card button {
  background-color: #4caf50;
  color: white;
  border: none;
  padding: 8px;
  margin-top: 10px;
  cursor: pointer;
  border-radius: 4px;
}

/* CART */
.cart {
  border: 1px solid #ccc;
  border-radius: 6px;
  padding: 15px;
  width: 280px;
}

.cart ul {
  list-style: none;
  padding: 0;
}

.cart li {
  display: flex;
  justify-content: space-between;
  margin-bottom: 10px;
}

.remove {
  background-color: #e53935;
  color: white;
  border: none;
  padding: 4px 8px;
  cursor: pointer;
  border-radius: 4px;
}

.empty {
  color: gray;
}
```

Output:


Smart Shopping Cart
Products
Headphones
₹1500
Add to Cart
Smart Watch
₹2500
Add to Cart
Bluetooth Speaker
₹3000
Add to Cart
Cart Summary
Your cart is empty


Smart Shopping Cart
Products
Headphones
₹1500
Add to Cart
Smart Watch
₹2500
Add to Cart
Bluetooth Speaker
₹3000
Add to Cart
Cart Summary
Headphones – ₹1500
Remove
Smart Watch – ₹2500
Remove
Bluetooth Speaker – ₹3000
Remove
Total: ₹7000


Smart Shopping Cart
Products
Headphones
₹1500
Add to Cart
Smart Watch
₹2500
Add to Cart
Bluetooth Speaker
₹3000
Add to Cart
Cart Summary
Headphones – ₹1500
Remove
Total: ₹1500

3. Internship Application Form by Implementing Event Handling and Form Validation in React.js

Program Coding:

App.jsx :

```
import InternshipApplication from
"./Components/InternshipApplication";

function App() {
  return <InternshipApplication />;
}

export default App;
```

InternshipApllication.jsx :

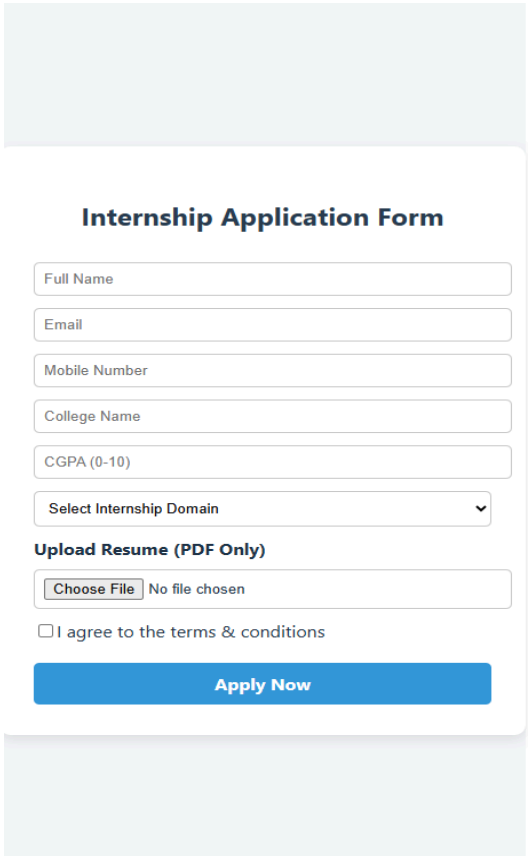
```
import React, { useState } from "react";

function InternshipApplication() {
  const [formData, setFormData] =
useState({
    fullName: "",
    email: "",
    mobile: "",
    college: "",
    cgpa: "",
    domain: "",
    resume: null,
    agree: false
  });

  const [errors, setErrors] = useState({});
  const [submitted, setSubmitted] =
useState(false);

  const handleChange = (e) => {
    const { name, value, type, checked, files }
= e.target;

    setFormData({
      ...formData,
      [name]:
        type === "checkbox"
          ? checked
          : type === "file"
          ? files[0]
          : value
    });
  };

  const validate = () => {
    let newErrors = {};

    if (!formData.fullName ||
formData.fullName.length < 3)
      newErrors.fullName = "Full Name must
be at least 3 characters";

    if (!/\S+@\S+\.\S+/.test(formData.email))
      newErrors.email = "Invalid email
format";

    if (!/^\d{10}$/.test(formData.mobile))
      newErrors.mobile = "Mobile number
must be 10 digits";

    if (!formData.college)
      newErrors.college = "College name
required";

    if (formData.cgpa < 0 || formData.cgpa >
10)
      newErrors.cgpa = "CGPA must be
between 0 and 10";
```

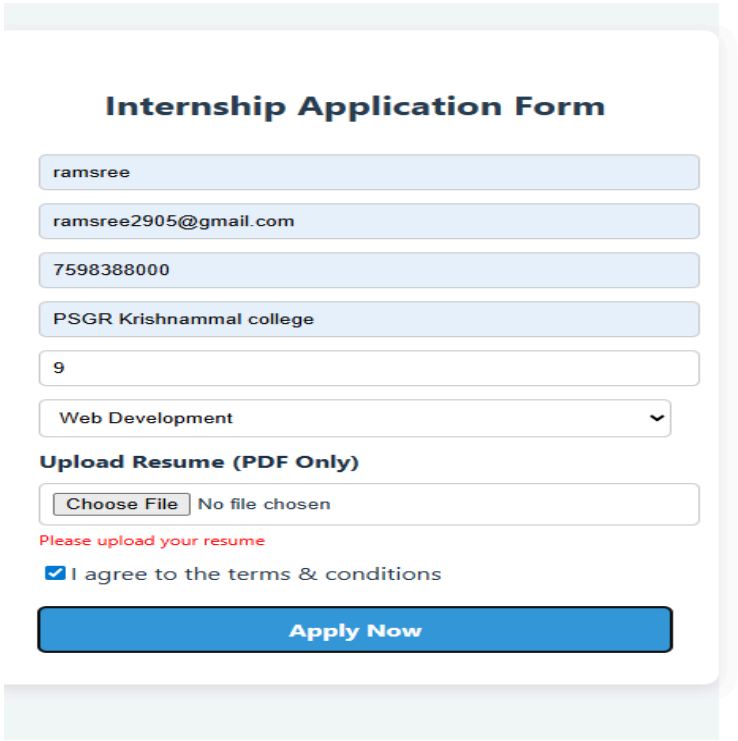
```
    if (!formData.domain)
      newErrors.domain = "Select internship
domain";

    if (!formData.resume)
      newErrors.resume = "Please upload your
resume";

    if (!formData.agree)
      newErrors.agree = "You must accept
terms";

    return newErrors;
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    const validationErrors = validate();
    setErrors(validationErrors);

    if (Object.keys(validationErrors).length
=== 0) {
      setSubmitted(true);
      alert("🎉 Application Submitted
Successfully!");
    }
  };

  return (
    <div style={styles.page}>
      <div style={styles.card}>
        <h2 style={styles.heading}>Internship
Application Form</h2>

        <form onSubmit={handleSubmit}>

          <input style={styles.input}
name="fullName" placeholder="Full Name"
onChange={handleChange} />
          <p
style={styles.error}>{errors.fullName}</p>

          <input style={styles.input}
name="email" placeholder="Email"
onChange={handleChange} />
          <p
style={styles.error}>{errors.email}</p>

          <input style={styles.input}
name="mobile" placeholder="Mobile
Number" onChange={handleChange} />
          <p
style={styles.error}>{errors.mobile}</p>

          <input style={styles.input}
name="college" placeholder="College
Name" onChange={handleChange} />
          <p
style={styles.error}>{errors.college}</p>

          <input style={styles.input}
type="number" name="cgpa"
placeholder="CGPA (0-10)"
onChange={handleChange} />
          <p
style={styles.error}>{errors.cgpa}</p>

          <select style={styles.input}
name="domain"
onChange={handleChange}>
            <option value="">Select Internship
Domain</option>
            <option value="Web">Web
Development</option>
            <option value="Data Science">Data
Science</option>
            <option value="AI">Artificial
Intelligence</option>
            <option value="Cyber
Security">Cyber Security</option>
```

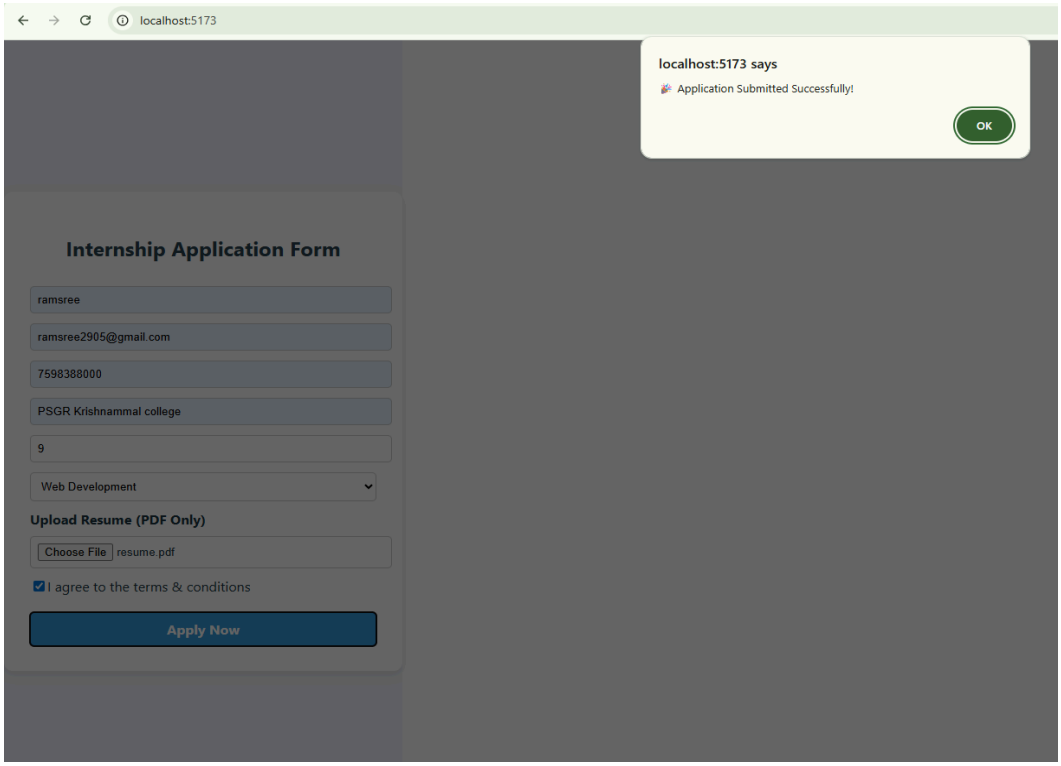
```
          <option value="Cloud
Computing">Cloud Computing</option>
          <option value="Mobile
App">Mobile App Development</option>
          <option value="UI/UX">UI/UX
Design</option>
          <option
value="DevOps">DevOps</option>
        </select>
        <p
style={styles.error}>{errors.domain}</p>

        <label style={styles.label}>Upload
Resume (PDF Only)</label>
        <input style={styles.input}
type="file" name="resume"
onChange={handleChange} />
        <p
style={styles.error}>{errors.resume}</p>

        <label style={styles.checkboxLabel}>
          <input type="checkbox"
name="agree" onChange={handleChange}
/>
          I agree to the terms & conditions
        </label>
        <p
style={styles.error}>{errors.agree}</p>

        <button style={styles.button}
type="submit">
          Apply Now
        </button>
      </form>

      {/* Optional: You can remove this
since alert is shown */}
      {submitted && (
        <p style={styles.success}>
          🎉 Application Submitted
Successfully!
        </p>
      )}
    </div>
  </div>
 );
}

const styles = {
 page: {
  backgroundColor: "#f4f6f9",
  minHeight: "100vh",
  display: "flex",
  justifyContent: "center",
  alignItems: "center"
 },
 card: {
  backgroundColor: "#ffffff",
  padding: "30px",
  borderRadius: "10px",
  boxShadow: "0 4px 12px rgba(0,0,0,0.1)",
  width: "400px"
 },
 heading: {
  textAlign: "center",
  color: "#2c3e50",
  marginBottom: "20px"
 },
 input: {
  width: "100%",
  padding: "8px",
  marginTop: "8px",
  borderRadius: "5px",
  border: "1px solid #ccc"
 },
 label: {
  marginTop: "10px",
  fontWeight: "bold",
  display: "block"
 },
 checkboxLabel: {
  marginTop: "10px",
```

```
  display: "block"
},
button: {
  width: "100%",
  padding: "10px",
  marginTop: "15px",
  backgroundColor: "#3498db",
  color: "white",
  border: "none",
  borderRadius: "5px",
  cursor: "pointer",
  fontWeight: "bold"
},
error: {
  color: "red",
  fontSize: "12px",
  margin: "4px 0"
},
success: {
  color: "green",
  textAlign: "center",
  marginTop: "15px",
  fontWeight: "bold"
}
};

export default InternshipApplication;
```

Output:


Internship Application Form
Full Name
Email
Mobile Number
College Name
CGPA (0-10)
Select Internship Domain
Upload Resume (PDF Only)
Choose File
No file chosen
I agree to the terms & conditions
Apply Now


Internship Application Form
ramsree
ramsree2905@gmail.com
7598388000
PSGR Krishnammal college
9
Web Development
Upload Resume (PDF Only)
Choose File
No file chosen
Please upload your resume
I agree to the terms & conditions
Apply Now


localhost:5173
localhost:5173 says
Application Submitted Successfully!
OK
Internship Application Form
ramsree
ramsree2905@gmail.com
7598388000
PSGR Krishnammal college
9
Web Development
Upload Resume (PDF Only)
Choose File
resume.pdf
I agree to the terms & conditions
Apply Now

4. Student Dashboard Using React Router with Multi-page Applications

Program Coding:

StudentDashboard.jsx

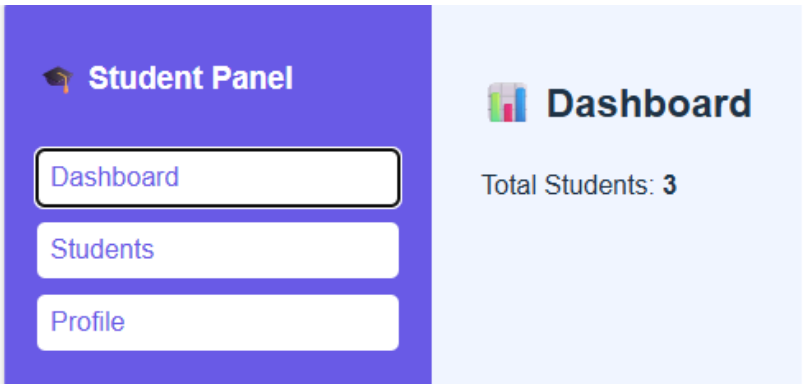
```
import { useState } from "react";

const StudentDashboard = () => {
  const [page, setPage] =
useState("dashboard");

  const students = [
    { id: 1, name: "Pavi", dept: "CSE", year:
"III" },
    { id: 2, name: "Nisha", dept: "IT", year:
"II" },
    { id: 3, name: "Swathi", dept: "ECE",
year: "I" },
  ];

  const renderContent = () => {
    if (page === "dashboard") {
      return (
        <div>
          <h2>📊 Dashboard</h2>
          <p>Total Students:
<strong>{students.length}</strong></p>
        </div>
      );
    }

    if (page === "students") {
      return (
        <div>
          <h2>📋 Student List</h2>
          <table style={styles.table}>
            <thead>
              <tr>
                <th style={styles.th}>Name</th>
                <th
style={styles.th}>Department</th>
                <th style={styles.th}>Year</th>
              </tr>
            </thead>
            <tbody>
              {students.map((s) => (
                <tr key={s.id}>
                  <td
style={styles.td}>{s.name}</td>
                  <td
style={styles.td}>{s.dept}</td>
                  <td
style={styles.td}>{s.year}</td>
                </tr>
              ))}
            </tbody>
          </table>
        </div>
      );
    }

    if (page === "profile") {
      return (
        <div>
          <h2>👤 Admin Profile</h2>
          <p><strong>Name:</strong>
Admin</p>
          <p><strong>Role:</strong> Student
Manager</p>
        </div>
      );
    }
  };

  return (
    <div style={styles.container}>
```

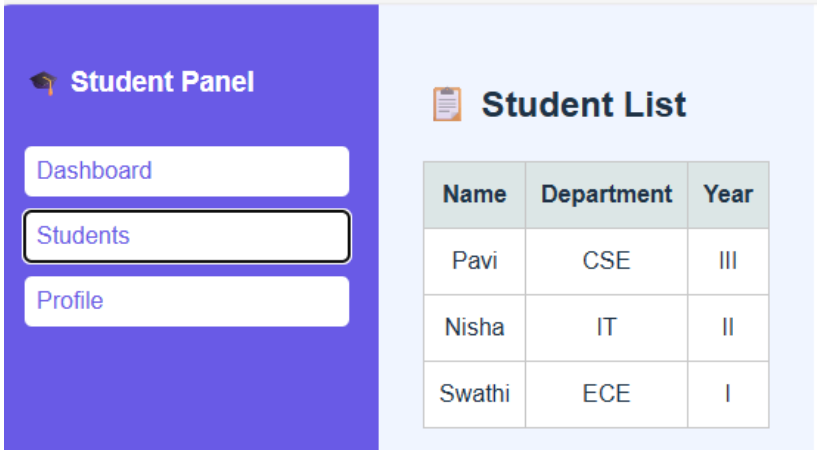
```
      {/* Navigation */}
      <div style={styles.sidebar}>
        <h3>🎓 Student Panel</h3>
        <button style={styles.navBtn}
onClick={() => setPage("dashboard")}>
          Dashboard
        </button>
        <button style={styles.navBtn}
onClick={() => setPage("students")}>
          Students
        </button>
        <button style={styles.navBtn}
onClick={() => setPage("profile")}>
          Profile
        </button>
      </div>

      {/* Content Area */}
      <div
style={styles.content}>{renderContent()}</
div>
    </div>
  );
};

const styles = {
  container: {
    minHeight: "100vh",
    display: "flex",
    fontFamily: "Arial, sans-serif",
    backgroundColor: "#f4f6ff",
  },
  sidebar: {
    width: "220px",
    backgroundColor: "#6c5ce7",
    color: "white",
    padding: "20px",
    display: "flex",
    flexDirection: "column",
    gap: "10px",
  },
  navBtn: {
    backgroundColor: "white",
    color: "#6c5ce7",
    border: "none",
    padding: "8px",
    borderRadius: "5px",
    cursor: "pointer",
    textAlign: "left",
  },
  content: {
    flex: 1,
    padding: "30px",
    backgroundColor: "#f4f6ff",
  },
  table: {
    width: "100%",
    borderCollapse: "collapse",
    marginTop: "15px",
    backgroundColor: "white",
  },
  th: {
    border: "1px solid #ccc",
    padding: "10px",
    backgroundColor: "#dfe6e9",
  },
  td: {
    border: "1px solid #ccc",
    padding: "10px",
    textAlign: "center",
  },
};

export default StudentDashboard;
```

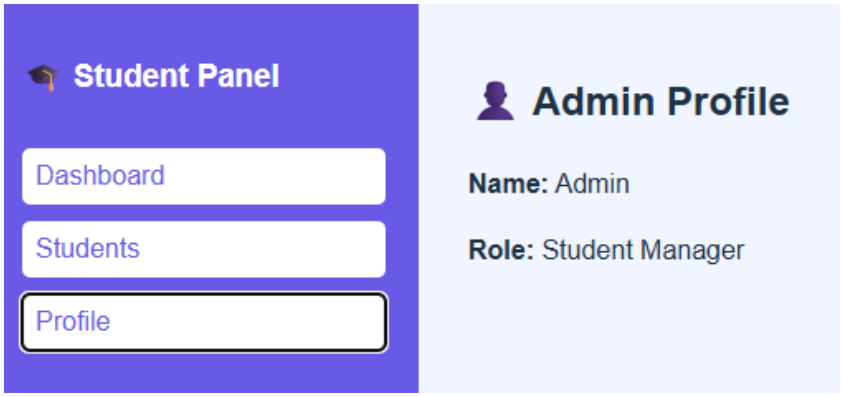
App.jsx:

```
import StudentDashboard from
"./FullStack/React/StudentDashboard"
function App() {
  return<StudentDashboard/>
}
export default Ap
```

Output:


Student Panel
Dashboard
Students
Profile
Dashboard
Total Students: 3


Student Panel
Dashboard
Students
Profile
Student List
Name
Department
Year
Pavi
CSE
III
Nisha
IT
II
Swathi
ECE
I


Student Panel
Dashboard
Students
Profile
Admin Profile
Name: Admin
Role: Student Manager

5. Smart Task Tracker Application Using React and Redux

Terminal:

npm install redux react-redux @reduxjs/toolkit

Program Coding:

store.js:

```
import { createStore, applyMiddleware } from "redux";
import { taskReducer } from "./reducer/taskReducer.js";
import logger from "./middleware/logger";

const store = createStore(taskReducer, applyMiddleware(logger));

export default store;
```

actions/taskActions.js

```
export const addTask = (text) => {
 return {
  type: "ADD_TASK",
  payload: text,
      };
 };

 export const deleteTask = (id) => {
  return {
  type: "DELETE_TASK",
  payload: id,
 };
};

export const toggleTask = (id) => {
 return {
  type: "TOGGLE_TASK",
  payload: id,
 };
};
```

reducers/taskReducer.js

```
const initialState = {
 tasks: [],
};

export const taskReducer = (state = initialState, action) => {
 switch (action.type) {
  case "ADD_TASK":
   return {
    ...state,
    tasks: [
     ...state.tasks,
     { id: Date.now(), text: action.payload, completed: false },
    ],
   };

  case "DELETE_TASK":
   return {
    ...state,
    tasks: state.tasks.filter(task => task.id !== action.payload),
   };

  case "TOGGLE_TASK":
   return {
    ...state,
    tasks: state.tasks.map(task =>
     task.id === action.payload
      ? { ...task, completed: !task.completed }
      : task
    ),
   };
```

```
    default:
      return state;
  }
};
```

middleware/logger.js

```
const logger = store => next => action => {
  console.log("Dispatching:", action);
  let result = next(action);
  console.log("Next State:", store.getState());
  return result;
};

export default logger;
```

index.js(vite means main.jsx)

```
import React from "react";
import ReactDOM from "react-dom/client";
import { Provider } from "react-redux";
import App from "./App";
import store from "./store";

const root =
ReactDOM.createRoot(document.getElemen
tById("root"));

root.render(
  <Provider store={store}>
    <App />
  </Provider>
);
```

components/AddTask.js:

```
import React, { useState } from "react";
import { useDispatch } from "react-redux";
import { addTask } from
"../Actions/taskActions";

function AddTask() {
  const [text, setText] = useState("");
  const dispatch = useDispatch();

  const handleAdd = () => {
    if (text.trim() === "") return;
    dispatch(addTask(text));
    setText("");
  };

  return (
    <div
      style={{
        display: "flex",
        gap: "10px",
        marginBottom: "25px"
      }}
    >
      <input
        type="text"
        placeholder="Enter your task..."
        value={text}
        onChange={(e) =>
setText(e.target.value)}
        style={{
          flex: 1,
          padding: "12px",
          borderRadius: "10px",
          border: "none",
          outline: "none",
          fontSize: "16px"
        }}
      />

      <button
        onClick={handleAdd}
        style={{
          padding: "12px 20px",
          borderRadius: "10px",
          border: "none",
```

```
        background: "linear-gradient(45deg,
#ff6b6b, #ff9f43)",
        color: "#fff",
        fontWeight: "bold",
        cursor: "pointer",
        transition: "0.3s"
       }}
     >
       Add
     </button>
   </div>
  );
}

export default AddTask;
```

components/TaskList.js:

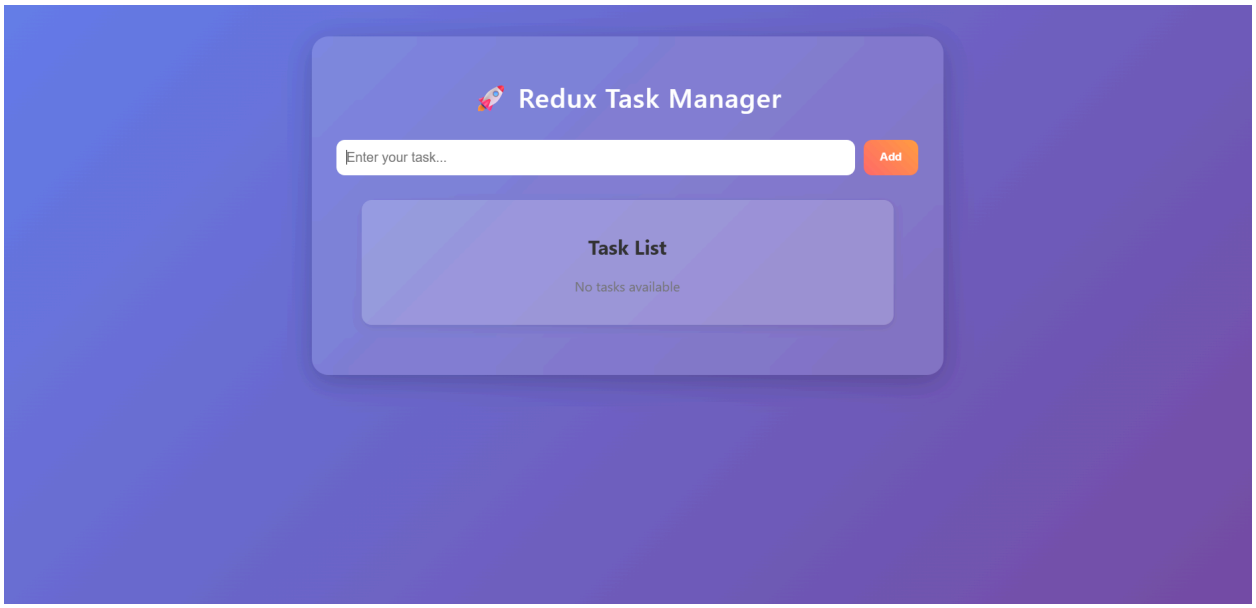
```
 import React from "react";
import { useSelector, useDispatch } from
"react-redux";
import { deleteTask, toggleTask } from
"../Actions/taskActions";

function TaskList() {
  const tasks = useSelector((state) =>
state.tasks);
  const dispatch = useDispatch();

  return (
   <div
    style={{
      maxWidth: "600px",
      margin: "30px auto",
      padding: "20px",

backgroundColor:"rgba(255,255,255,0.2)",
           color: "#fff",

      borderRadius: "12px",
      boxShadow: "0 4px 12px
rgba(0,0,0,0.08)",
      fontFamily: "Segoe UI, sans-serif"
    }}
   >
    <h2
     style={{
       textAlign: "center",
       marginBottom: "20px",
       color: "#333"
     }}
    >
     Task List
    </h2>

    {tasks.length === 0 ? (
     <p style={{ textAlign: "center", color:
"#777" }}>
       No tasks available
     </p>
    ) : (
     tasks.map((task) => (
      <div
       key={task.id}
       style={{
         display: "flex",
         justifyContent: "space-between",
         alignItems: "center",
         padding: "12px",
         marginBottom: "10px",
         backgroundColor: "#f9fafc",
         borderRadius: "8px",
         border: "1px solid #e5e7eb"
       }}
      >
       <span
        onClick={() =>
dispatch(toggleTask(task.id))}
        style={{
          textDecoration: task.completed ?
"line-through" : "none",
```

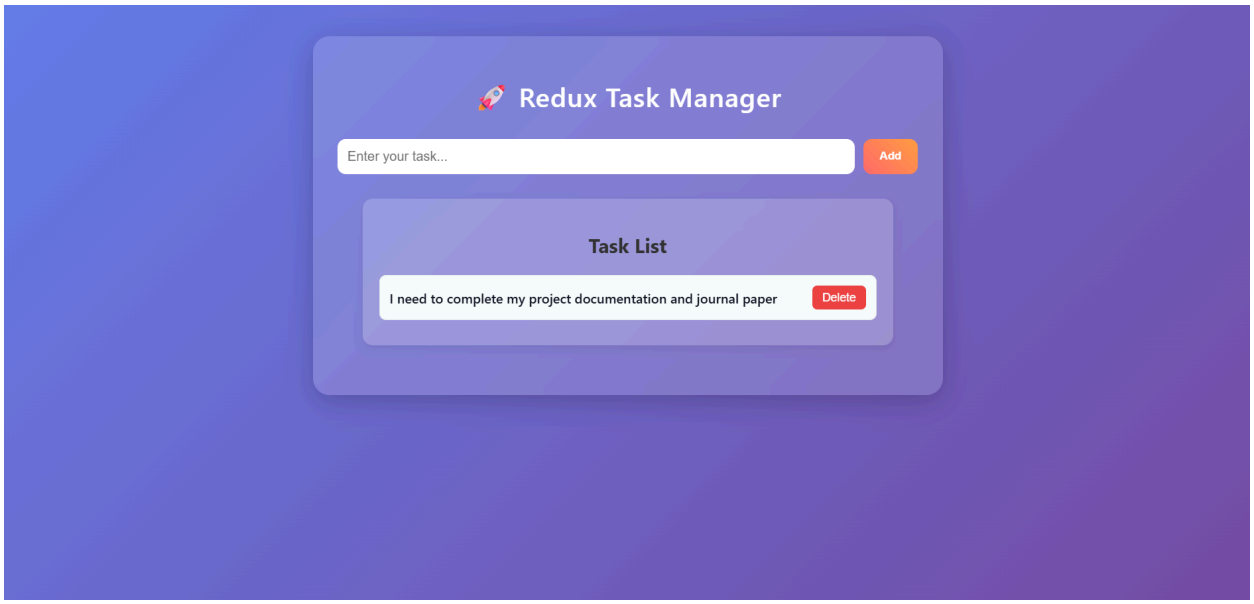
```
            cursor: "pointer",
            color: task.completed ? "#9ca3af"
: "#111827",
            fontWeight: "500"
           }}
          >
           {task.text}
          </span>
          <button
           onClick={() =>
dispatch(deleteTask(task.id))}
           style={{
            padding: "6px 12px",
            backgroundColor: "#ef4444",
            color: "#fff",
            border: "none",
            borderRadius: "6px",
            cursor: "pointer",
            fontSize: "14px"
           }}
          >
           Delete
          </button>
         </div>
        ))
      )}
    </div>
  );
}

export default TaskList;
```

app.js:

```
 import React from "react";
import AddTask from
"./components/AddTask";
import TaskList from
"./components/TaskList";

function App() {
  return (
    <div
     style={{
      minHeight: "100vh",
      background: "linear-gradient(135deg,
#667eea, #764ba2)",
      padding: "40px 20px",
      fontFamily: "Segoe UI, sans-serif"
     }}
    >
     <div
      style={{
       maxWidth: "700px",
       margin: "auto",
       background: "rgba(255, 255, 255,
0.15)",
       backdropFilter: "blur(15px)",
       padding: "30px",
       borderRadius: "20px",
       boxShadow: "0 8px 32px
rgba(0,0,0,0.2)",
       color: "#fff"
      }}
     >
      <h1
       style={{
        textAlign: "center",
        marginBottom: "30px",
        fontWeight: "600",
        letterSpacing: "1px"
       }}
      >
       🚀 Redux Task Manager
      </h1>

      <AddTask />
      <TaskList />
     </div>
    </div>
  );
}
export default App;
```

Output:


Redux Task Manager
Enter your task...
Add
Task List
No tasks available


Redux Task Manager
Enter your task...
Add
Task List
I need to complete my project documentation and journal paper
Delete

6. Online Job Application Form Using Angular

Program Coding:

app.config.ts

```
import { ApplicationConfig } from
'@angular/core';
import { provideRouter } from
'@angular/router';
import { routes } from './app.routes';

export const appConfig: ApplicationConfig
= {
  providers: [provideRouter(routes)]
};
```

app.routes.ts

```
import { Routes } from '@angular/router';
import { App } from './app';

export const routes: Routes =
  { path: '', component: App }
];
```

app.html

```
<div class="container">
  <h2>Online Job Application</h2>
  <p class="mandatory-note">
  <span class="star">*</span> indicates
mandatory fields
</p>
  <form #jobForm="ngForm"
(ngSubmit)="submitForm(jobForm)">
    <!-- Personal Details -->
    <h3>Personal Details</h3>
    <label>Full Name <span
class="star">*</span></label>
<input
  type="text"
  name="name"
  [(ngModel)]="applicant.name"
  required
  #name="ngModel"
  [class.invalid]="name.invalid &&
name.touched">
<div class="error" *ngIf="name.invalid &&
name.touched">
  Name is required
</div>
   <label>Email <span
class="star">*</span></label>
<input
  type="text"
  name="email"
  [(ngModel)]="applicant.email"
  required

pattern="^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.
-]+\.[a-zA-Z]{2,}$"
  #email="ngModel"
  [class.invalid]="email.invalid &&
email.touched">
<div class="error"
*ngIf="email.errors?.['required'] &&
email.touched">
  Email is required
</div>
<div class="error"
*ngIf="email.errors?.['pattern'] &&
email.touched">
  Enter a valid email address (example:
name123@gmail.com)
</div>
    <label>Phone Number <span
class="star">*</span></label>
<input
  type="text"
```

```
  name="phone"
  [(ngModel)]="applicant.phone"
  pattern="[0-9]{10}"
  required
  #phone="ngModel"
  [class.invalid]="phone.invalid &&
phone.touched">
<div class="error" *ngIf="phone.invalid
&& phone.touched">
  Enter valid 10-digit phone number
</div>
    <label>Address <span
class="star">*</span></label>
    <textarea name="address"
[(ngModel)]="applicant.address"
required></textarea>
    <!-- Job Details -->
    <h3>Job Details</h3>
    <label>Position Applying For <span
class="star">*</span></label>
    <select name="position"
[(ngModel)]="applicant.position" required>
      <option value="">Select</option>
      <option *ngFor="let p of
positions">{{p}}</option>
    </select>
    <label>Preferred Location <span
class="star">*</span></label>
    <select name="location"
[(ngModel)]="applicant.location" required>
      <option value="">Select</option>
      <option *ngFor="let l of
locations">{{l}}</option>
    </select>
    <label>Expected Salary <span
class="star">*</span></label>
    <input type="number" name="salary"
[(ngModel)]="applicant.salary" required>
    <!-- Education -->
    <h3>Educational Qualification</h3>
    <label>Highest Qualification <span
class="star">*</span></label>
    <select name="qualification"
[(ngModel)]="applicant.qualification"
required>
      <option value="">Select</option>
      <option *ngFor="let q of
qualifications">{{q}}</option>
    </select>
    <label>University / College <span
class="star">*</span></label>
    <select name="university"
[(ngModel)]="applicant.university"
required>
      <option value="">Select</option>
      <option *ngFor="let u of
universities">{{u}}</option>
    </select>
    <label>Year of Passing <span
class="star">*</span></label>
    <select name="year"
[(ngModel)]="applicant.year" required>
      <option value="">Select</option>
      <option *ngFor="let y of
years">{{y}}</option>
    </select>
    <!-- Skills -->
    <h3>Skills <span
class="star">*</span></h3>
<div class="checkbox-group" *ngFor="let
skill of skillsList">
  <label>
    <input
      type="checkbox"
      [value]="skill"
      (change)="onSkillChange($event)">
    {{skill}}
  </label>
</div>
    <!-- Experience -->
```

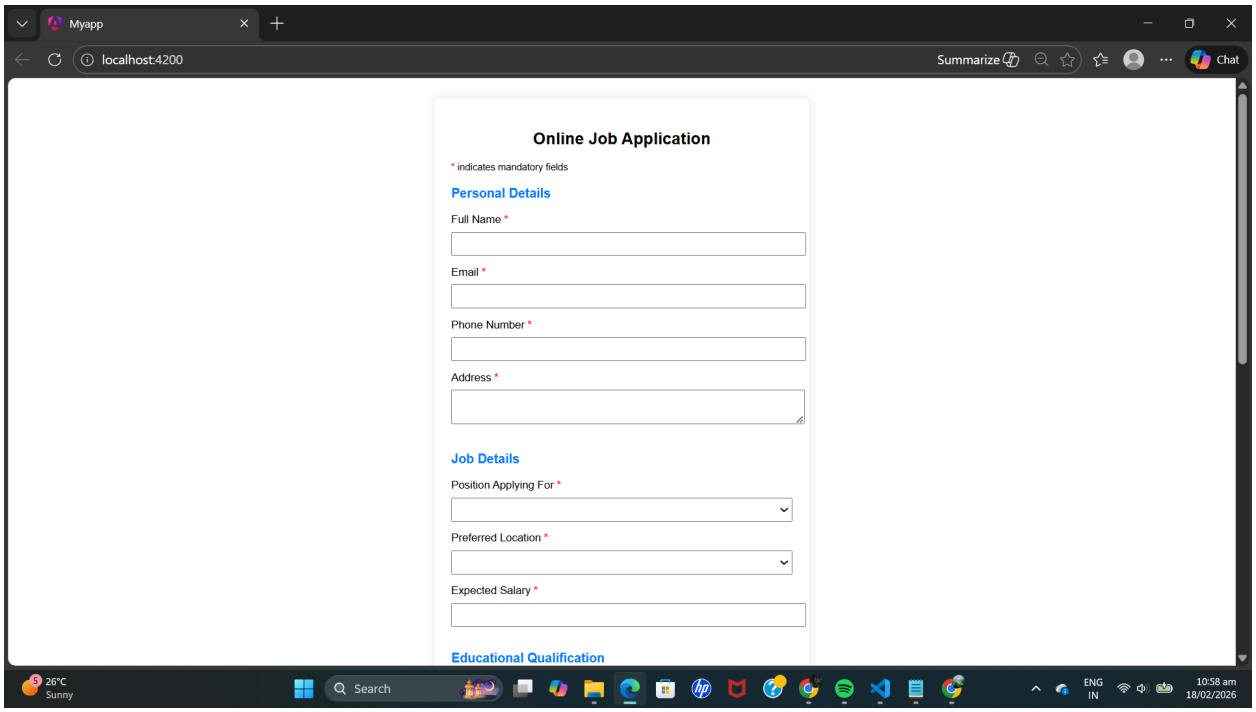
```
    <h3>Experience <span
class="star">*</span></h3>
<label class="radio-group">
  <input type="radio" name="exp"
value="Fresher"
[(ngModel)]="applicant.exp">
  Fresher
</label>
<label class="radio-group">
  <input type="radio" name="exp"
value="Experienced"
[(ngModel)]="applicant.exp">
  Experienced
</label>
<div *ngIf="applicant.exp ===
'Experienced'">
  <label>Years of Experience</label>
  <input type="number" name="yearsExp"
[(ngModel)]="applicant.yearsExp">
</div>
    <!-- Resume Upload -->
    <h3>Upload Resume <span
class="star">*</span></h3>
    <input type="file">
    <!-- Declaration -->
   <div class="agree">
  <input
    type="checkbox"
    name="agree"
    [(ngModel)]="applicant.agree"
    required
    #agree="ngModel" >
  <span>
    I confirm the above information is correct
    <span class="star">*</span>
  </span>
</div>
<div class="error" *ngIf="agree.invalid &&
agree.touched">
  You must accept before submitting
</div>
    <button type="submit"
[disabled]="jobForm.invalid">
      Submit Application
    </button>
  </form>
</div>
```

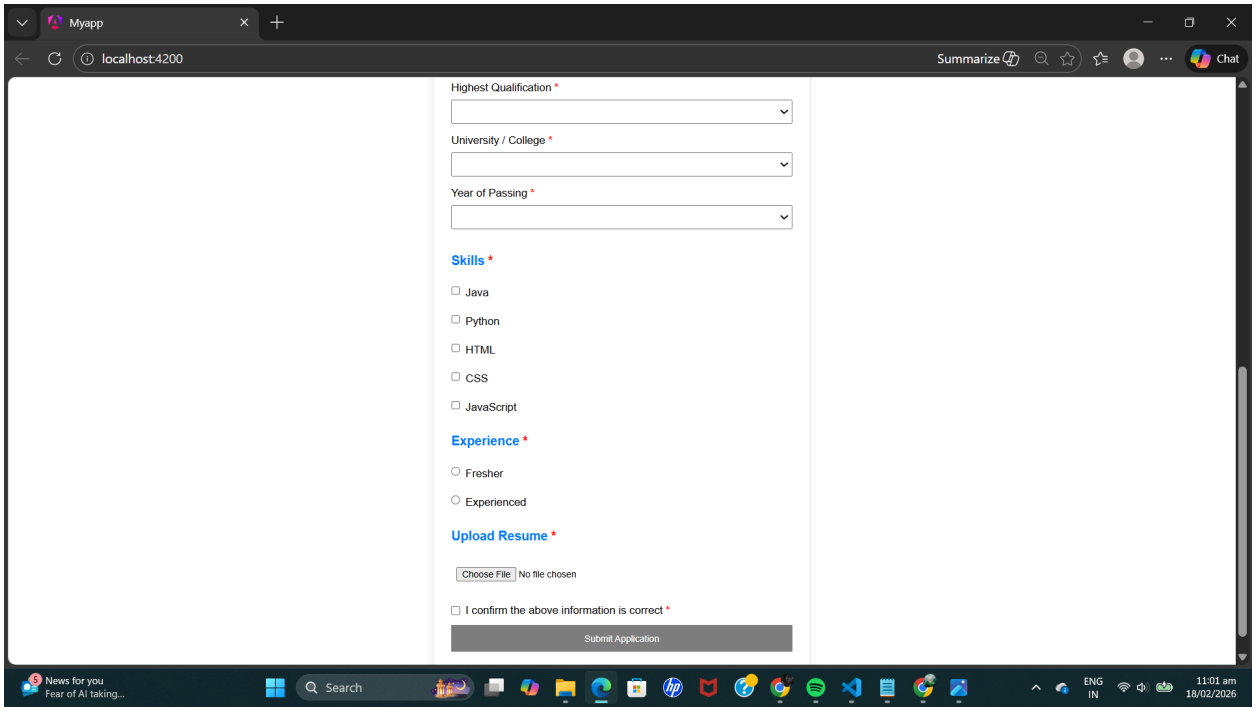
app.css

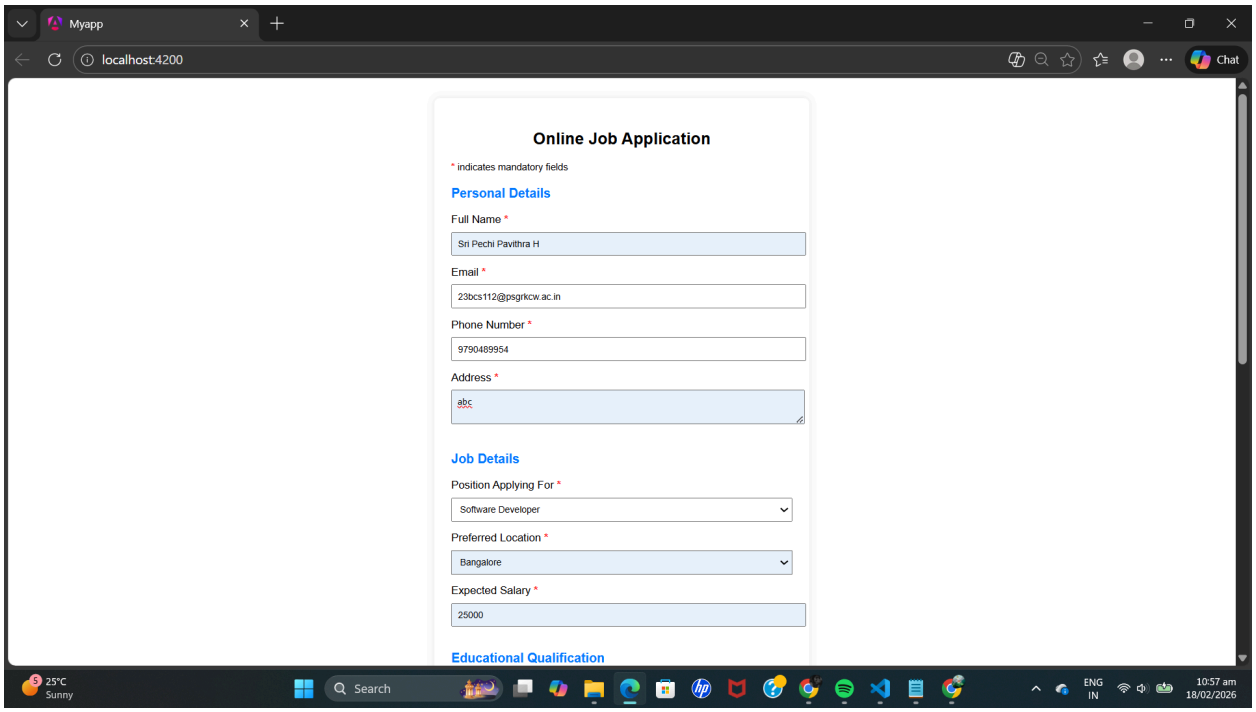
```
.container {
  width: 500px;
  margin: 30px auto;
  background: white;
  padding: 25px;
  border-radius: 8px;
  box-shadow: 0 0 10px rgba(0,0,0,0.1);
  font-family: Arial;
}
h2 {
  text-align: center;
}
h3 {
  margin-top: 20px;
  color: #007BFF;
}
input, select, textarea {
  width: 100%;
  padding: 8px;
  margin: 8px 0 15px 0;
}
button {
  width: 100%;
  padding: 12px;
  background: #007BFF;
  color: white;
  border: none;
}
button:disabled {
  background: gray;
}
.checkbox-group label,
.radio-group {
```

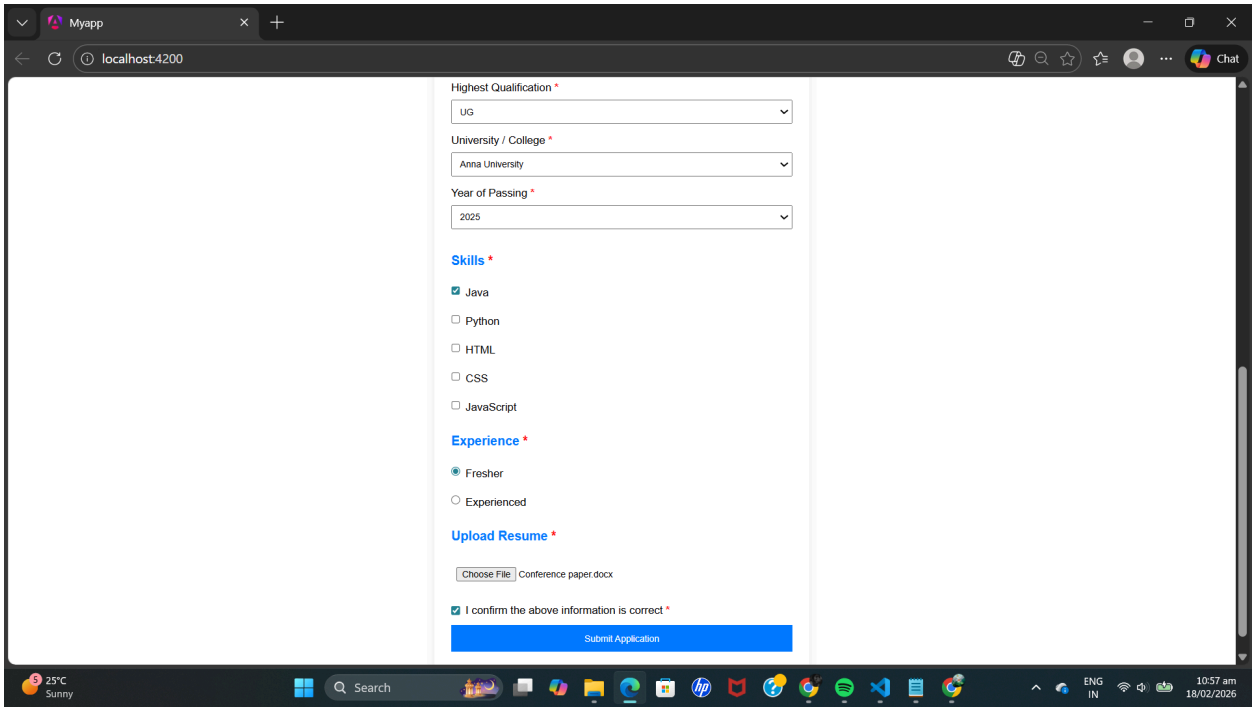
```
  display: flex;
  align-items: center;
  gap: 8px;
  margin-bottom: 6px;
}
.checkbox-group input,
.radio-group input {
  width: auto;
}
.star {
  color: red;
  font-weight: bold;
}
.mandatory-note {
  margin-bottom: 15px;
  font-size: 14px;
}
.error {
  color: red;
  font-size: 12px;
  margin-top: -10px;
  margin-bottom: 10px;
}
.invalid {
  border: 2px solid red;
}
.agree {
  display: flex;
  align-items: flex-start;
  gap: 8px;
  margin-top: 10px;
}
.agree input {
  margin-top: 3px;
  width: auto;
}
```

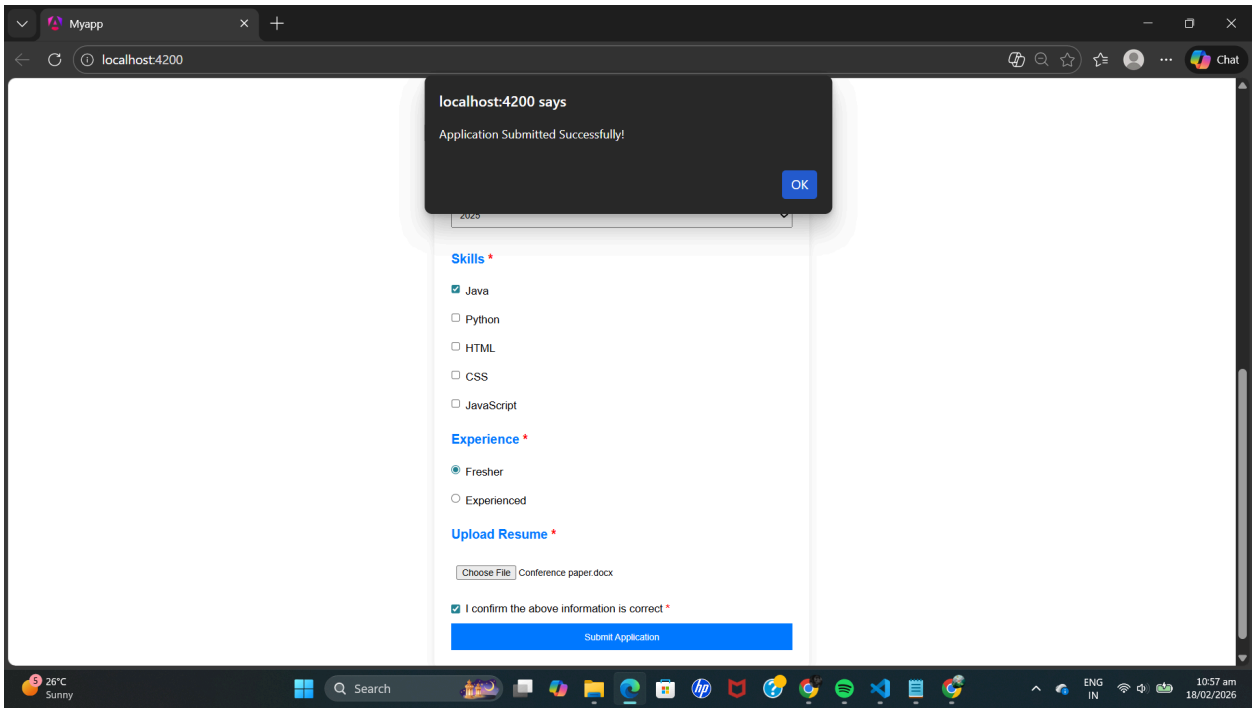
OUTPUT:


Myapp
localhost:4200
Summarize
Chat
Online Job Application
* indicates mandatory fields
Personal Details
Full Name *
Email *
Phone Number *
Address *
Job Details
Position Applying For *
Preferred Location *
Expected Salary *
Educational Qualification


Myapp
localhost:4200
Summarize
Chat
Highest Qualification *
University / College *
Year of Passing *
Skills *
Java
Python
HTML
CSS
JavaScript
Experience *
Fresher
Experienced
Upload Resume *
Choose File
No file chosen
I confirm the above information is correct *
Submit Application


Myapp
localhost:4200
Chat
Online Job Application
* indicates mandatory fields
Personal Details
Full Name *
Sri Pechi Pavithra H
Email *
23bcs112@psgrkcw.ac.in
Phone Number *
9790489954
Address *
abc
Job Details
Position Applying For *
Software Developer
Preferred Location *
Bangalore
Expected Salary *
25000
Educational Qualification
25°C
Sunny
Search
ENG
IN
10:57 am
18/02/2026


Myapp
localhost:4200
Chat
Highest Qualification *
UG
University / College *
Anna University
Year of Passing *
2025
Skills *
Java
Python
HTML
CSS
JavaScript
Experience *
Fresher
Experienced
Upload Resume *
Choose File
Conference paper.docx
I confirm the above information is correct *
Submit Application
25°C
Sunny
Search
ENG
IN
10:57 am
18/02/2026


Myapp
localhost:4200
Chat
localhost:4200 says
Application Submitted Successfully!
OK
Skills *
Java
Python
HTML
CSS
JavaScript
Experience *
Fresher
Experienced
Upload Resume *
Choose File
Conference paper.docx
I confirm the above information is correct *
Submit Application
26°C
Sunny
Search
ENG
IN
10:57 am
18/02/2026

7. Hospital Management System for Patients using AngularJS Services and REST API

Program Coding:

Index.html:

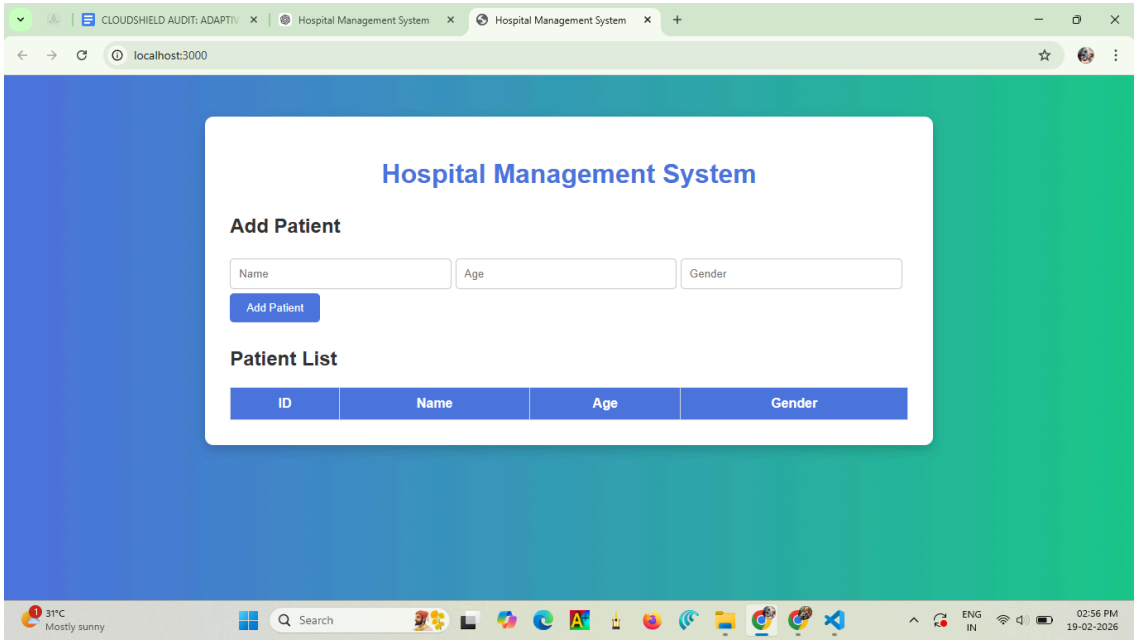
```
<!DOCTYPE html>
<html ng-app="app">
<head>
  <title>Hospital Management System</title>
  <script src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></script>

  <style>
    body {
      font-family: Arial, Helvetica, sans-serif;
      background: linear-gradient(to right, #4e73df, #1cc88a);
      margin: 0;
      padding: 0;
    }

    .container {
      width: 60%;
      margin: 50px auto;
      background: white;
      padding: 30px;
      border-radius: 10px;
      box-shadow: 0 5px 15px rgba(0,0,0,0.2);
    }

    h1 {
      text-align: center;
      color: #4e73df;
    }
    h2 {
      margin-top: 30px;
      color: #333;
    }

    input {
      padding: 10px;
      margin: 5px 0;
      width: 30%;
      border-radius: 5px;
      border: 1px solid #ccc;
    }

    button {
      padding: 10px 20px;
      background-color: #4e73df;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
    }

    button:hover {
      background-color: #2e59d9;
    }

    table {
      width: 100%;
      margin-top: 15px;
      border-collapse: collapse;
    }

    table, th, td {
      border: 1px solid #ddd;
```

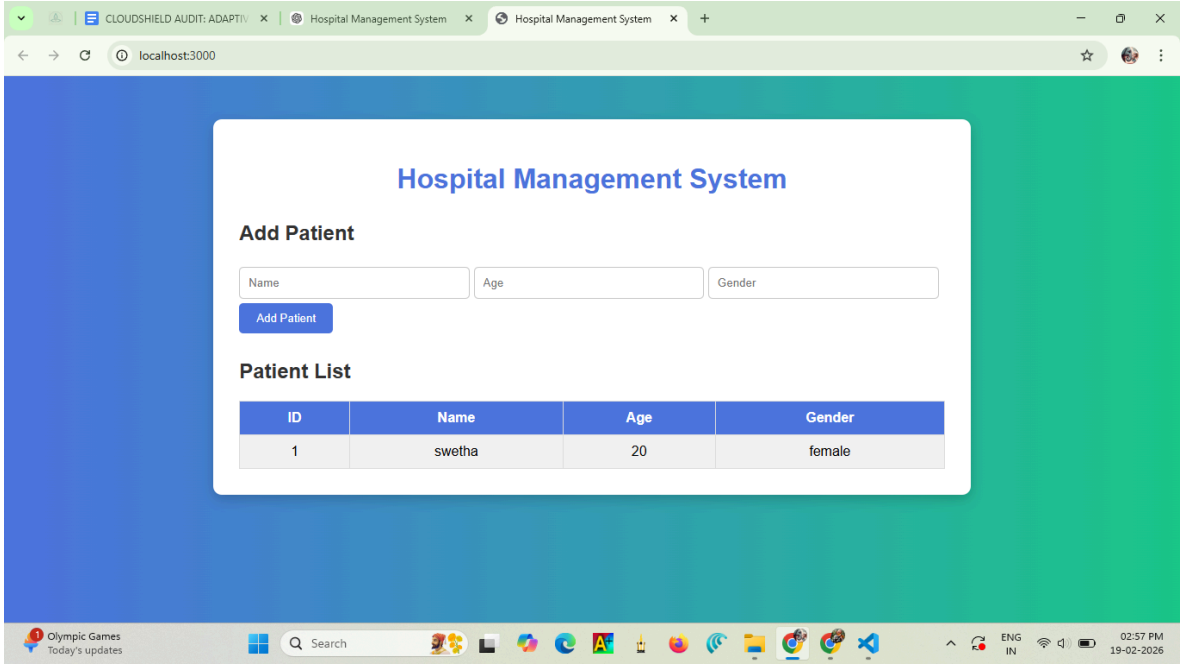
```
        }

        th {
            background-color: #4e73df;
            color: white;
            padding: 10px;
        }

        td {
            padding: 10px;
            text-align: center;
        }

        tr:nth-child(even) {
            background-color: #f2f2f2;
        }

    </style>
</head>

<body ng-controller="ctrl">

<div class="container">
    <h1>Hospital Management System</h1>

    <h2>Add Patient</h2>

    <input ng-model="p.name"
placeholder="Name">
    <input ng-model="p.age"
placeholder="Age">
    <input ng-model="p.gender"
placeholder="Gender">
    <button ng-click="add()">Add
Patient</button>

    <h2>Patient List</h2>

    <table>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Age</th>
            <th>Gender</th>
        </tr>
        <tr ng-repeat="x in patients">
            <td>{{x.id}}</td>
            <td>{{x.name}}</td>
            <td>{{x.age}}</td>
            <td>{{x.gender}}</td>
        </tr>
    </table>
</div>

<script>
var app = angular.module("app", []);

app.controller("ctrl", function($scope, $http)
{

    function load(){

$http.get("/patients").then(function(res){
            $scope.patients = res.data;
        });
    }

    $scope.add = function(){
        $http.post("/patients",
$scope.p).then(function(){
            $scope.p = {};
            load();
        });
    };

    load();
});
</script>

</body>
</html>
```

Server.js

```
const express = require("express");
const app = express();

app.use(express.json());
app.use(express.static(__dirname));

let patients = [];
let id = 1;

// Get all patients
app.get("/patients", (req, res) => {
   res.json(patients);
});

// Add patient
app.post("/patients", (req, res) => {
   const patient = {
      id: id++,
      name: req.body.name,
      age: req.body.age,
      gender: req.body.gender
   };

   patients.push(patient);
   res.json(patient);
});

app.listen(3000, () => {
   console.log("Server running at
http://localhost:3000");
});
```

Output:


localhost:3000
Hospital Management System
Add Patient
Name
Age
Gender
Add Patient
Patient List
ID
Name
Age
Gender


localhost:3000
Hospital Management System
Add Patient
Name
Age
Gender
Add Patient
Patient List
ID
Name
Age
Gender
1
swetha
20
female

8.Simple Attendance Tracking System using Node.js: Installation and HTTP Server Setup

Program Coding:

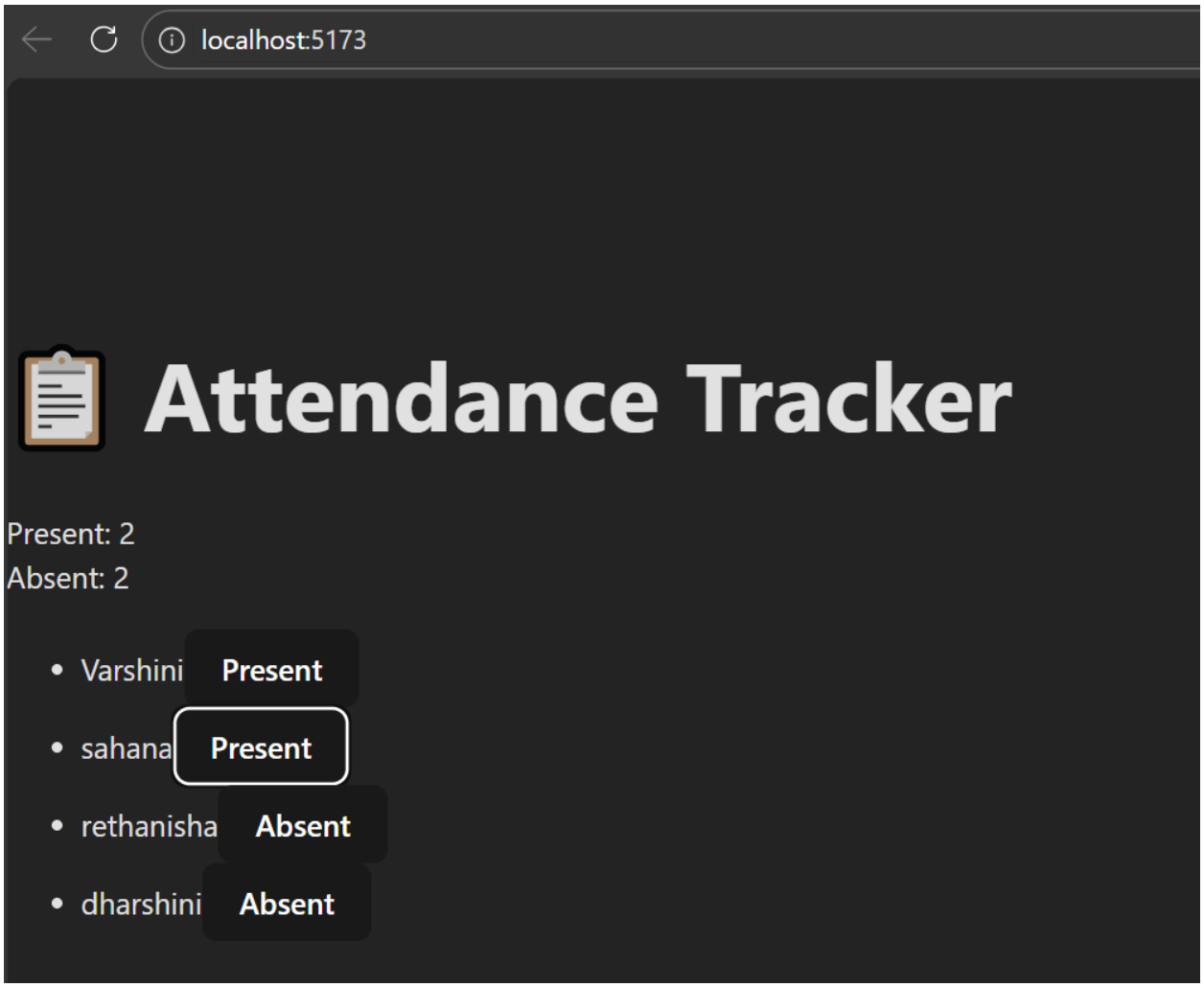
App.jsx:

```
import Attendance from "./Attendance";
function App() {
  return <Attendance/>;
}
export default App;
```

Attendance.jsx:

```
import { useState } from "react";
function Attendance() {
  const [students, setStudents] = useState([
    { name: "Varshini", present: false },
    { name: "sahana", present: false },
    { name: "rethanisha", present: false },
     ]);
    const toggleAttendance = (index) => {
    const updatedStudents = [...students];
    updatedStudents[index].present =
!updatedStudents[index].present;
    setStudents(updatedStudents);
  };

  const presentCount = students.filter(s =>
s.present).length;
  const absentCount = students.length -
presentCount;

  return (
    <div className="app">
      <div className="card">
        <h1>📋 Attendance Tracker</h1>

        <div className="stats">
          <div className="present">Present:
{presentCount}</div>
          <div className="absent">Absent:
{absentCount}</div>
        </div>

        <ul>
          {students.map((student, index) => (
            <li key={index}>
              <span>{student.name}</span>
              <button
                className={student.present ?
"btn-present" : "btn-absent"}
                onClick={() =>
toggleAttendance(index)}
              >
                {student.present ? "Present" :
"Absent"}
              </button>
            </li>
          ))}
        </ul>
      </div>
    </div>
  );
}
export default Attendance;
```

Output:


localhost:5173
Attendance Tracker
Present: 2
Absent: 2
Varshini Present
sahana Present
rethanisha Absent
dharshini Absent

9. Hospital Appointment RESTful API using Express.js: routing, middleware and Error Handling

Program Coding:

index.js:

```
const express = require("express");
const appointmentRoutes =
require("./routes/appointmentRoutes");
const logger =
require("./middleware/logger");

const app = express();
const PORT = 3000;

// Middleware
app.use(express.json()); // parses JSON
app.use(logger);         // logs requests

// Routes
app.use("/api", appointmentRoutes);

// Error handling
app.use((err, req, res, next) => {
  res.status(500).json({ error: err.message });
});

// Start server
app.listen(PORT, () => {
  console.log(`Hospital Appointment API
running on port ${PORT}`);
});
```

//Middleware

logger.js:

```
function logger(req, res, next) {
  console.log(`${req.method} ${req.url}`);
  next();
}

module.exports = logger;
```

//ROUTING

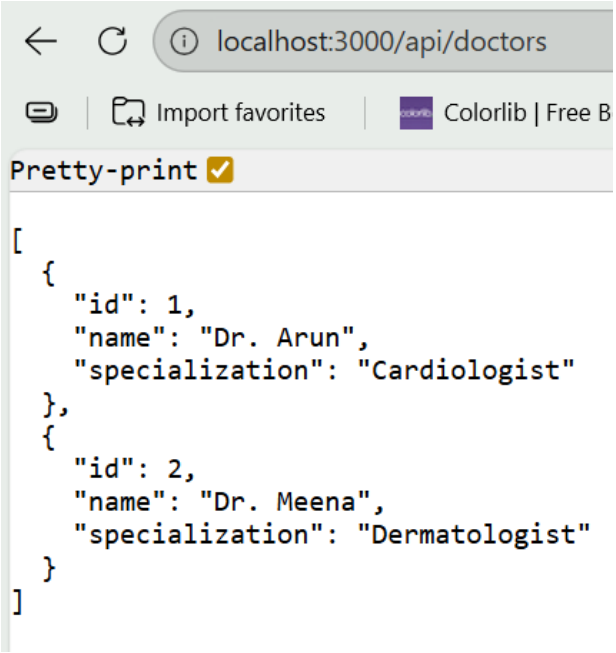
appointmentRoutes.js:

```
const express = require("express");
const router = express.Router();

// Sample doctors
let doctors = [
  { id: 1, name: "Dr. Arun", specialization:
"Cardiologist" },
  { id: 2, name: "Dr. Meena", specialization:
"Dermatologist" }
];

// Pre-filled appointments
let appointments = [
  { id: 1, patientName: "Alice", doctorId: 1
},
  { id: 2, patientName: "Bob", doctorId: 2 }
];

// GET doctors
router.get("/doctors", (req, res) =>
res.json(doctors));

// GET appointments
router.get("/appointments", (req, res) =>
res.json(appointments));

// POST appointment (optional, still works)
router.post("/appointments", (req, res) => {
  const { patientName, doctorId } =
req.body;
  if (!patientName || !doctorId)
    return res.status(400).json({ error:
"Patient name and doctor ID are required"
});
```

```
  const appointment = { id:
appointments.length + 1, patientName,
doctorId };
  appointments.push(appointment);
  res.status(201).json({ message:
"Appointment booked successfully",
appointment });
});

// DELETE appointment (optional, still
works)
router.delete("/appointments/:id", (req, res)
=> {
  const id = parseInt(req.params.id);
  const initialLength = appointments.length;
  appointments = appointments.filter(app =>
app.id !== id);

  if (appointments.length === initialLength)
    return res.status(404).json({ error:
"Appointment not found" });

  res.json({ message: "Appointment
cancelled" });
});

module.exports = router;
```

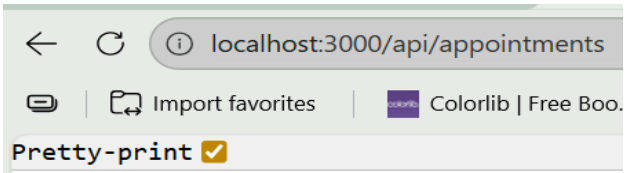
Output:

GET Method: /api/doctors



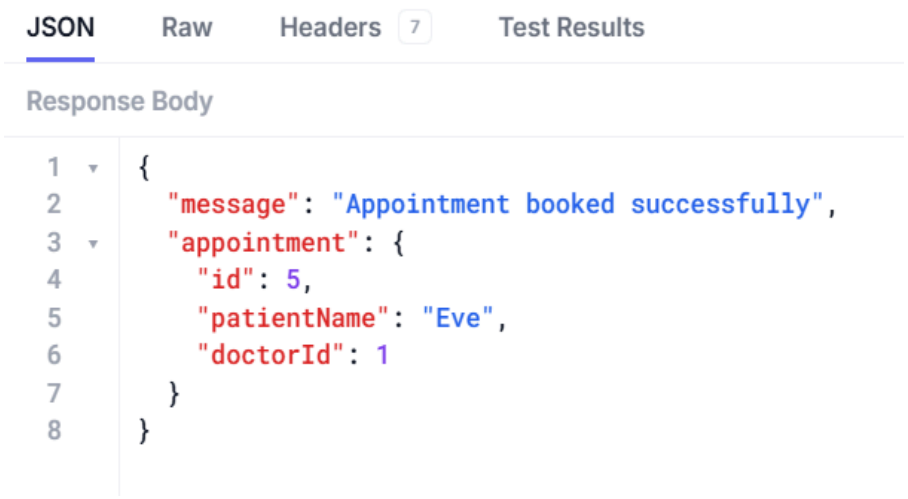

```
[
  {
    "id": 1,
    "name": "Dr. Arun",
    "specialization": "Cardiologist"
  },
  {
    "id": 2,
    "name": "Dr. Meena",
    "specialization": "Dermatologist"
  }
]
```

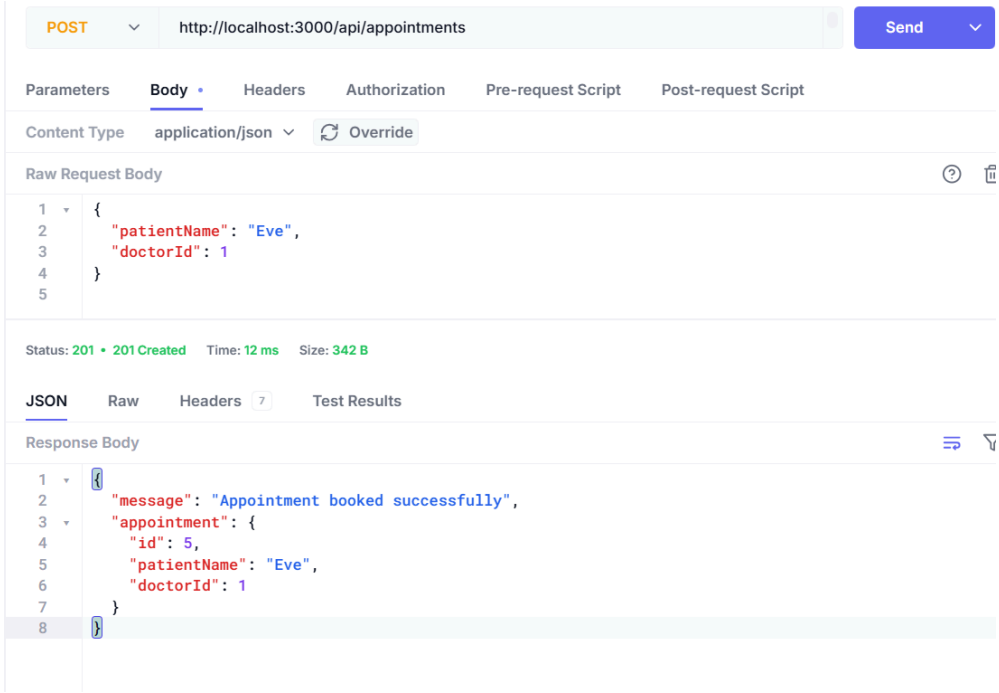

GET Method: /api/appointments



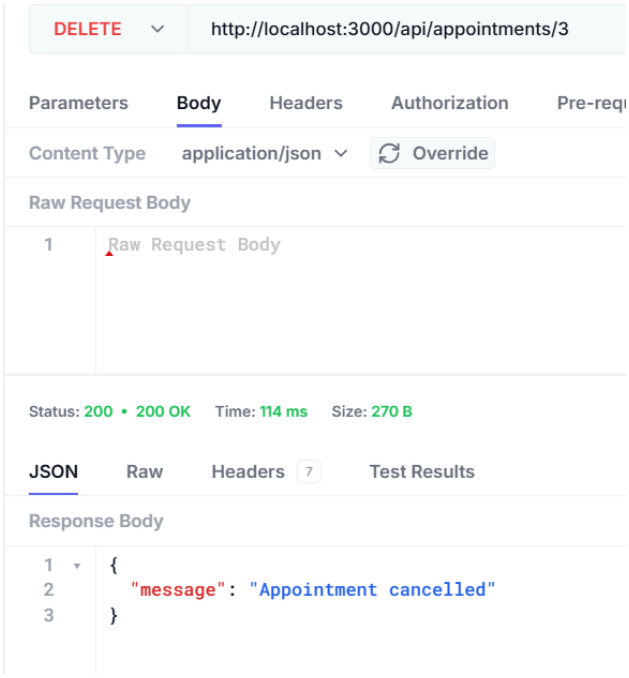

```
[
  {
    "id": 1,
    "patientName": "Alice",
    "doctorId": 1
  },
  {
    "id": 2,
    "patientName": "Bob",
    "doctorId": 2
  },
  {
    "id": 3,
    "patientName": "Charlie",
    "doctorId": 1
  },
  {
    "id": 4,
    "patientName": "Diana",
    "doctorId": 2
  }
]
```

POST Method:


JSON
Raw
Headers 7
Test Results
Response Body
{
"message": "Appointment booked successfully",
"appointment": {
"id": 5,
"patientName": "Eve",
"doctorId": 1
}
}


POST
http://localhost:3000/api/appointments
Send
Parameters
Body
Headers
Authorization
Pre-request Script
Post-request Script
Content Type
application/json
Override
Raw Request Body
{
"patientName": "Eve",
"doctorId": 1
}
Status: 201 • 201 Created
Time: 12 ms
Size: 342 B
JSON
Raw
Headers 7
Test Results
Response Body
{
"message": "Appointment booked successfully",
"appointment": {
"id": 5,
"patientName": "Eve",
"doctorId": 1
}
}

DELETE Method: /api/appointments/3



10. CRUD operations with MongoDB for Students Registration

Program code:

Backend:

config/db.js:

```
const mongoose = require("mongoose");
const connectDB = async () => {
  try {
    await
mongoose.connect("mongodb://127.0.0.1:27
017/authdb");
    console.log("MongoDB Connected");
  } catch (error) {
    console.error(error);
    process.exit(1);
  }
};
module.exports = connectDB;
```

models/User.js:

```
const mongoose = require("mongoose");
const userSchema = new
mongoose.Schema({
  name: String,
  email: String,
  password: String
});
module.exports = mongoose.model("User",
userSchema);
```

routes/auth.routes.js:

```
const express = require("express");
const router = express.Router();
const User = require("../models/User");
const jwt = require("jsonwebtoken");
// REGISTER
router.post("/register", async (req, res) => {
  const user = new User(req.body);
  await user.save();
  res.json({ message: "User registered
successfully" });
});

// LOGIN
router.post("/login", async (req, res) => {
  const user = await User.findOne({ email:
req.body.email });
if (!user) {
    return res.status(401).json({ message:
"Invalid user" });
  }

  const token = jwt.sign({ id: user._id },
"secretkey", {
    expiresIn: "1h",
  });

  res.json({ token });
});


//  VERIFY TOKEN API
router.get("/verify-token", (req, res) => {
  const authHeader =
req.headers.authorization;
  if (!authHeader) {
    return res.status(401).json({ message:
"Token missing" });
  }
  const token = authHeader.split(" ")[1];
  console.log("Auth Header:",
req.headers.authorization);
  console.log("Extracted Token:", token);
  try {
    const decoded = jwt.verify(token,
"secretkey");
    res.json({
      valid: true,
      userId: decoded.id,
    });
  } catch (err) {
    res.status(401).json({
      valid: false,
      message: "Invalid or expired token",
```

```
    });
  }
});
module.exports = router;
```

server.js:

```
const express = require("express");
const cors = require("cors");
const connectDB = require("./config/db");
const app = express();
app.use(cors());
app.use(express.json());
connectDB();
app.use("/api",
require("./routes/auth.routes"));
app.listen(5000, () => {
  console.log("Server running on port
5000");
});
```

Update package.json:

```
“scripts”:{
“test”: “echo \ “Error” no test specified\”
&& exit 1”,
“start” : “nodemon server.js”,
“dev”: “nodemon server.js”
}
```

Frontend:

pages\Auth.css:

```
.auth-container {
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background: linear-gradient(135deg,
#ff9f1c, #2ec4b6);
  font-family: "Segoe UI", Tahoma,
sans-serif;
}
/* Glass card */
.auth-card {
  background: rgba(255, 255, 255, 0.25);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border-radius: 16px;
  padding: 35px 40px;
  width: 100%;
  max-width: 380px;
  text-align: center;
  border: 1px solid rgba(255, 255, 255,
0.35);
  box-shadow: 0 8px 32px rgba(0, 0, 0,
0.25);
}
.auth-card h2 {
  margin-bottom: 20px;
  color: #1f2937;
  font-weight: 600;
  letter-spacing: 0.5px;
}
/* Inputs */
.auth-card input {
  width: 100%;
  padding: 12px 14px;
  margin-bottom: 16px;
  border-radius: 8px;
  border: none;
  outline: none;
  font-size: 14px;
  background: rgba(255, 255, 255, 0.7);
  color: #333;
}
.auth-card input::placeholder {
  color: #666;
}
.auth-card input:focus {
  box-shadow: 0 0 0 2px rgba(255, 159, 28,
0.6);
}
/* Button */
.auth-card button {
  width: 100%;
  padding: 12px;
  margin-top: 5px;
  background: linear-gradient(135deg,
#ff9f1c, #ffbf69);
  color: #fff;
  border: none;
  border-radius: 8px;
  font-size: 15px;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.3s ease;
}
.auth-card button:hover {
```

```
  transform: translateY(-2px);
  background: linear-gradient(135deg,
#ff8c00, #ffa94d);
  box-shadow: 0 6px 15px rgba(255, 159,
28, 0.5);
}
/* Links */
.switch-text {
  margin-top: 18px;
  font-size: 14px;
  color: #1f2937;
}
.switch-text a,
.link {
  color: #065f46;
  cursor: pointer;
  font-weight: 600;
  text-decoration: none;
}
.switch-text a:hover,
.link:hover {
  text-decoration: underline;
}
/* Error */
.error {
  color: #b91c1c;
  margin-bottom: 12px;
  font-size: 14px;
  font-weight: 500;
}
```

pages\Register.js:

```
import axios from "axios";
import { useState } from "react";
import { useNavigate, Link } from
"react-router-dom";
import "./Auth.css";
export default function Register() {
  const [name, setName] = useState("");
  const [email, setEmail] = useState("");
  const [password, setPassword] =
useState("");
  const [error, setError] = useState("");
  const navigate = useNavigate();
  const register = async () => {
    try {
      await
axios.post("http://localhost:5000/api/register
", {
        name,
        email,
        password,
      });
      alert("Registered successfully");
      navigate("/login");
    } catch (err) {
      setError("Registration failed. Try
again.");
    }
  };
  return (
    <div className="auth-container">
      <div className="auth-card">
        <h2>Create Account</h2>
        {error && <p
className="error">{error}</p>}
        <input
          placeholder="Full Name"
          value={name}
          onChange={e =>
setName(e.target.value)}
        />
        <input
          type="email"
          placeholder="Email"
          value={email}
          onChange={e =>
setEmail(e.target.value)}
        />
        <input
          type="password"
          placeholder="Password"
          value={password}
          onChange={e =>
setPassword(e.target.value)}
        />
        <button
onClick={register}>Register</button>
        <p className="switch-text">
          Already have an account?{" "}
          <Link to="/login">Login</Link>
        </p>
      </div>
```

```
    </div>
  );
}
```

pages\Login.js

```
import axios from "axios";
import { useState } from "react";
import { useNavigate, Link } from
"react-router-dom";
import "./Auth.css";

export default function Login() {
  const [email, setEmail] = useState("");
  const [password, setPassword] =
useState("");
  const [error, setError] = useState("");
  const navigate = useNavigate();
  const login = async () => {
    try {
      const res = await
axios.post("http://localhost:5000/api/login",
{
        email,
        password,
      });

      localStorage.setItem("token",
res.data.token);
      navigate("/"); // Home page
    } catch (err) {
      setError("Invalid email or password");
    }
  };
  return (
    <div className="auth-container">
      <div className="auth-card">
        <h2>Login</h2>
        {error && <p
className="error">{error}</p>}
        <input
          type="email"
          placeholder="Email"
          value={email}
          onChange={e =>
setEmail(e.target.value)}
        />
        <input
          type="password"
          placeholder="Password"
          value={password}
          onChange={e =>
setPassword(e.target.value)}
        />
        <button
onClick={login}>Login</button>
        <p className="switch-text">
          Don’t have an account?{" "}
          <Link to="/register">Register</Link>
        </p>
      </div>
    </div>
  );
}
```

pages\Home.css:

```
.home-container {
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  background: linear-gradient(135deg,
#ff9f1c, #2ec4b6);
  font-family: "Segoe UI", Tahoma,
sans-serif;
}
/* Glass card */
.home-card {
  background: rgba(255, 255, 255, 0.25);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  padding: 45px 40px;
  width: 100%;
  max-width: 440px;
  border-radius: 18px;
  text-align: center;
  border: 1px solid rgba(255, 255, 255,
0.35);
  box-shadow: 0 8px 32px rgba(0, 0, 0,
0.25);
}
.home-card h2 {
  margin-bottom: 12px;
  color: #1f2937;
  font-size: 26px;
  font-weight: 600;
```

```
}
.home-card p {
  color: #1f2937;
  margin-bottom: 30px;
  font-size: 15px;
  opacity: 0.9;
}
/* Logout button */
.home-card button {
  padding: 12px 32px;
  background: linear-gradient(135deg,
#ff9f1c, #ffbf69);
  color: #fff;
  border: none;
  border-radius: 10px;
  cursor: pointer;
  font-size: 15px;
  font-weight: 600;
  transition: all 0.3s ease;
}
.home-card button:hover {
  transform: translateY(-2px);
  background: linear-gradient(135deg,
#ff8c00, #ffa94d);
  box-shadow: 0 6px 15px rgba(255, 159,
28, 0.5);
}
```

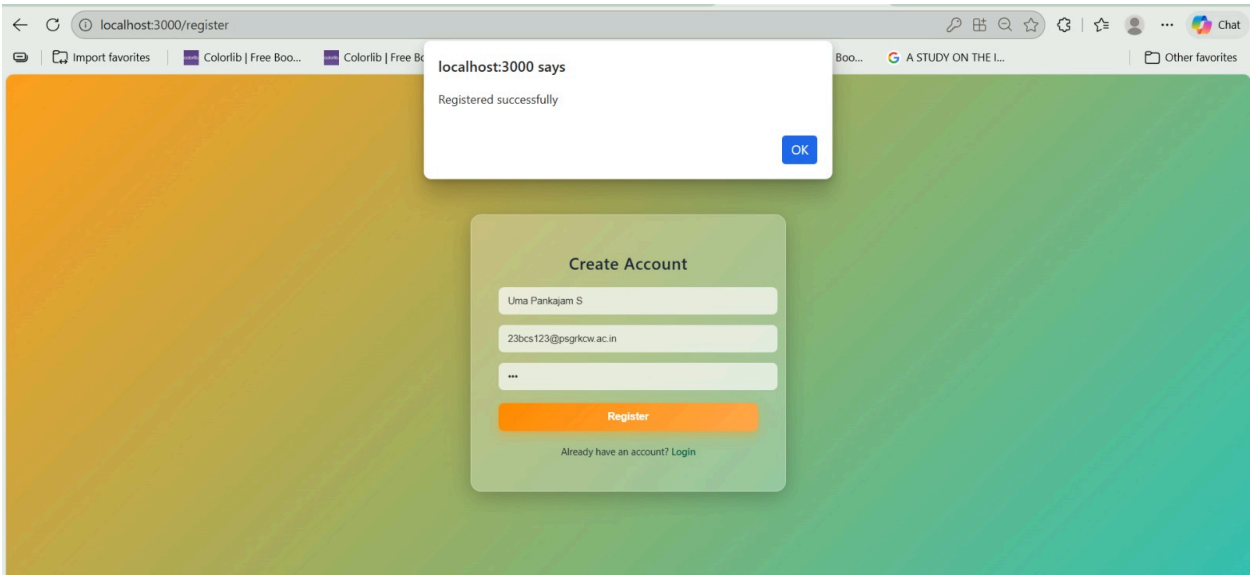
pages\Home.js:

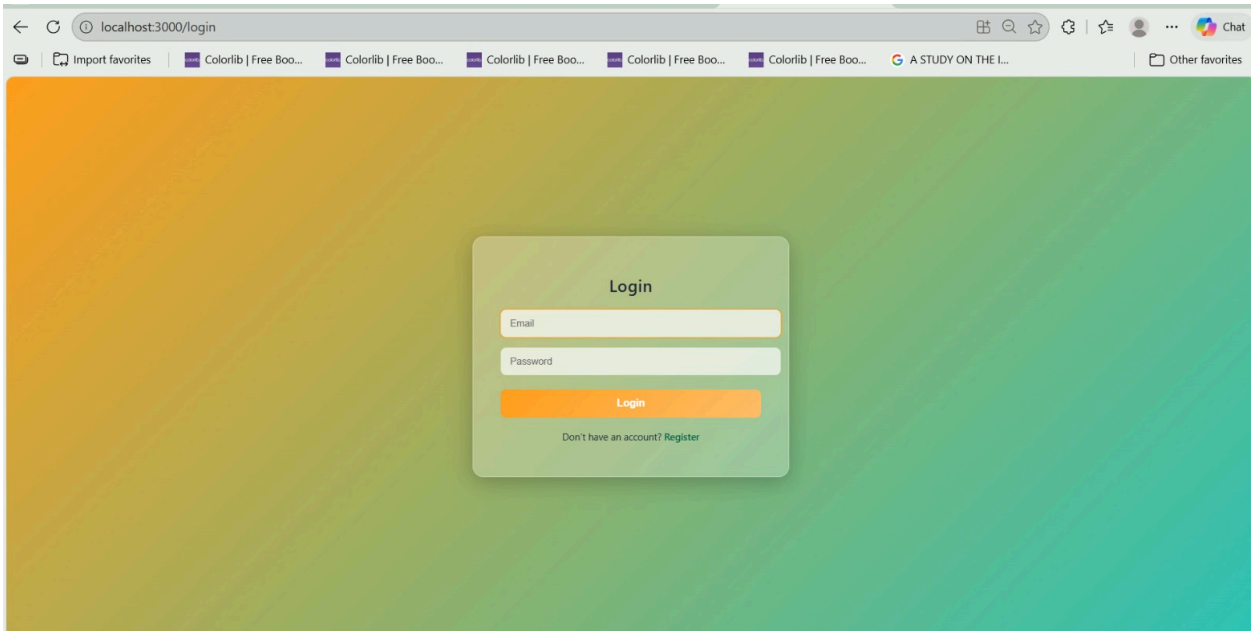
```
import axios from "axios";
import { useEffect } from "react";
import { useNavigate } from
"react-router-dom";
import "./Home.css";
export default function Home() {
  const navigate = useNavigate();
  useEffect(() => {
    const verify = async () => {
      try {
        const token=
localStorage.getItem("token")
        await
axios.get("http://localhost:5000/api/verify-to
ken", {
          headers: {
            Authorization: `Bearer ${token}`,
          },
        });
      } catch {
        localStorage.removeItem("token");
        navigate("/login");
      }
    };
    verify();
  }, [navigate]);
  const logout = () => {
    localStorage.removeItem("token");
    navigate("/login");
  };
  return (
    <div className="home-container">
      <div className="home-card">
        <h2>Welcome 👋</h2>
        <p>You are successfully logged
in.</p>
        <button
onClick={logout}>Logout</button>
      </div>
    </div>
  );
}
```

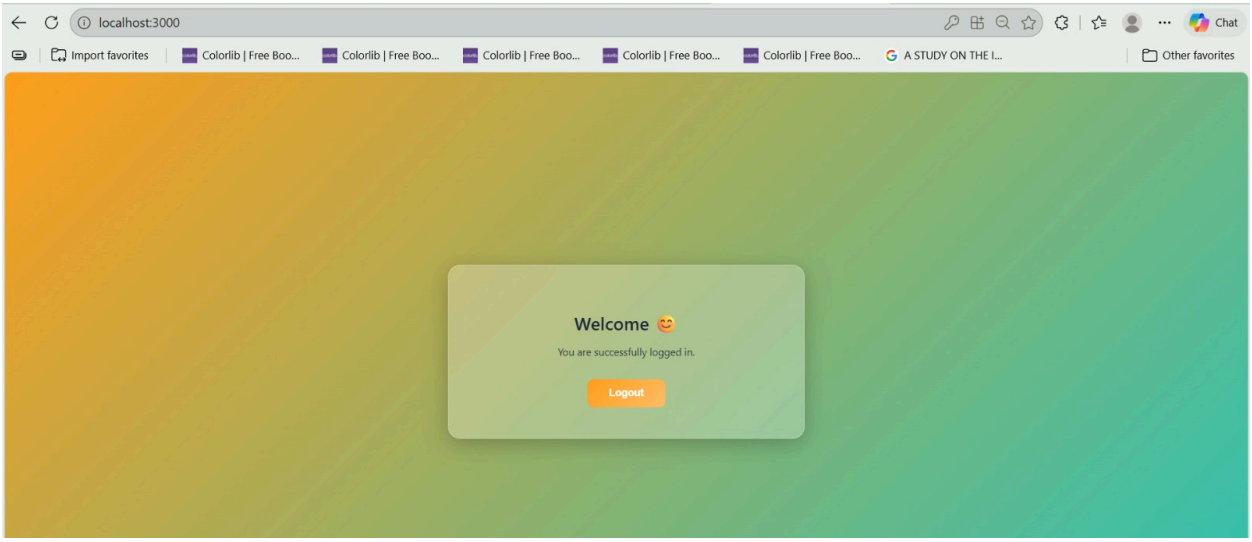
App.js:

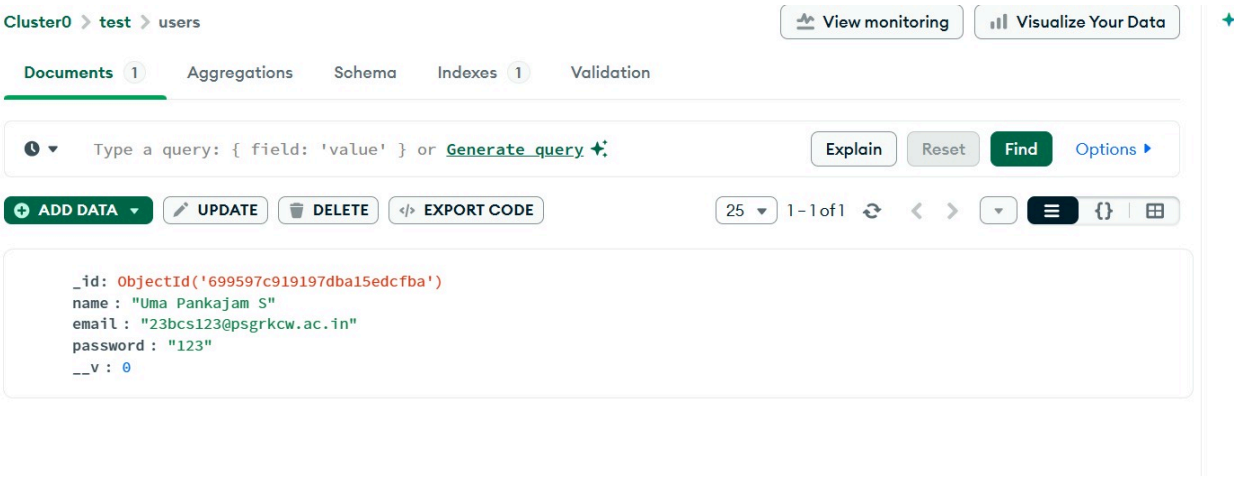
```
import { BrowserRouter, Routes, Route,
Navigate } from "react-router-dom";
import Login from "./pages/Login";
import Register from "./pages/Register";
import Home from "./pages/Home";
function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Home />}
/>
        <Route path="/login"
element={<Login />} />
        <Route path="/register"
element={<Register />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

Output:


localhost:3000/register
Import favorites
Colorlib | Free Boo...
A STUDY ON THE I...
Other favorites
Chat
localhost:3000 says
Registered successfully
OK
Create Account
Uma Pankajam S
23bcs123@psgrkcw.ac.in
Register
Already have an account? Login


localhost:3000/login
Import favorites
Colorlib | Free Boo...
A STUDY ON THE I...
Other favorites
Chat
Login
Email
Password
Login
Don't have an account? Register


localhost:3000
Chat
Import favorites
Colorlib | Free Boo...
Colorlib | Free Boo...
Colorlib | Free Boo...
Colorlib | Free Boo...
Colorlib | Free Boo...
A STUDY ON THE I...
Other favorites
Welcome
You are successfully logged in.
Logout


Cluster0 > test > users
View monitoring
Visualize Your Data
Documents 1
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT CODE
25
1 - 1 of 1
_id: ObjectId('699597c919197dba15edcfba')
name : "Uma Pankajam S"
email : "23bcs123@psgrkcw.ac.in"
password : "123"
__v : 0

11. Online Examination Portal: Authentication and Session Management Using JWT in Node.js Applications

Program Coding:

index.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Exam Portal</title>
  <style>
    body {
      font-family: Arial;
      background-color: #f2f2f2;
      text-align: center;
      margin-top: 40px;
    }
    .container {
      background: white;
      width: 400px;
      margin: auto;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0px 0px 10px gray;
    }
    input {
      width: 90%;
      padding: 8px;
      margin: 8px 0;
    }
    button {
      padding: 8px 15px;
      margin-top: 10px;
      cursor: pointer;
    }
    hr {
      margin: 20px 0;
    }
    #examSection {
      display: none;
      text-align: left;
    }
  </style>
</head>
<body
<div class="container">
<h2>Online Exam Portal</h2>
<!-- Register -->
<h3>Register</h3>
<input type="text" id="regUsername"
placeholder="Username"><br>
<input type="password" id="regPassword"
placeholder="Password"><br>
<button
onclick="register()">Register</button>
<hr
<!-- Login -->
<h3>Login</h3>
<input type="text" id="loginUsername"
placeholder="Username"><br>
<input type="password"
id="loginPassword"
placeholder="Password"><br>
<button onclick="login()">Login</button>
<p id="message"></p>
<hr>
<!-- Exam Section -->
<div id="examSection">
  <h3>Exam Questions</h3>
  <div id="questions"></div>
  <button onclick="submitExam()">Submit
Exam</button>
  <button
onclick="logout()">Logout</button>
</div>
</div>
<script>
const API_URL = "http://localhost:5000";
let correctAnswers = {};
// REGISTER
async function register() {
  const username =
document.getElementById("regUsername").
value;
  const password =
document.getElementById("regPassword").
value;
```

```
    const res = await fetch(API_URL +
"/register", {
        method: "POST",
        headers: { "Content-Type":
"application/json" },
        body: JSON.stringify({ username,
password })
    });
    const data = await res.json();

document.getElementById("message").inner
Text = data.message;
}
async function login() {
    const username =
document.getElementById("loginUsername
").value;
    const password =
document.getElementById("loginPassword"
).value;
    const res = await fetch(API_URL +
"/login", {
        method: "POST",
        headers: { "Content-Type":
"application/json" },
        body: JSON.stringify({ username,
password })
    });
    const data = await res.json();
    if (data.token) {
        localStorage.setItem("token",
data.token);

document.getElementById("message").inner
Text = "Login successful!";
        loadExam();
    } else {

document.getElementById("message").inner
Text = data.message;
    }
}
async function loadExam() {
    const token =
localStorage.getItem("token");
    const res = await fetch(API_URL +
"/exam", {
        method: "GET",
        headers: {
            "Authorization": "Bearer " + token
        }
    });
    const data = await res.json();
    if (data.exam) {

document.getElementById("examSection").
style.display = "block";
        let output = "";
        correctAnswers = {};
        data.exam.forEach(q => {
            correctAnswers[q.id] = q.answer;
            output += `<div>
                    <p><b>Q${q.id}:
${q.question}</b></p>`;
            q.options.forEach(option => {
                output += `
                  <label>
                    <input type="radio"
name="q${q.id}" value="${option}">
                      ${option}
                  </label><br>
                `;
            });
            output += `</div><br>`;
        });

document.getElementById("questions").inne
rHTML = output;
    } else {

document.getElementById("message").inner
Text = "Unauthorized!";
    }
}
function submitExam() {
    let score = 0;
    let total =
Object.keys(correctAnswers).length;

    for (let q in correctAnswers) {
        const selected =
document.querySelector(`input[name="q${q
}"]:checked`);
```

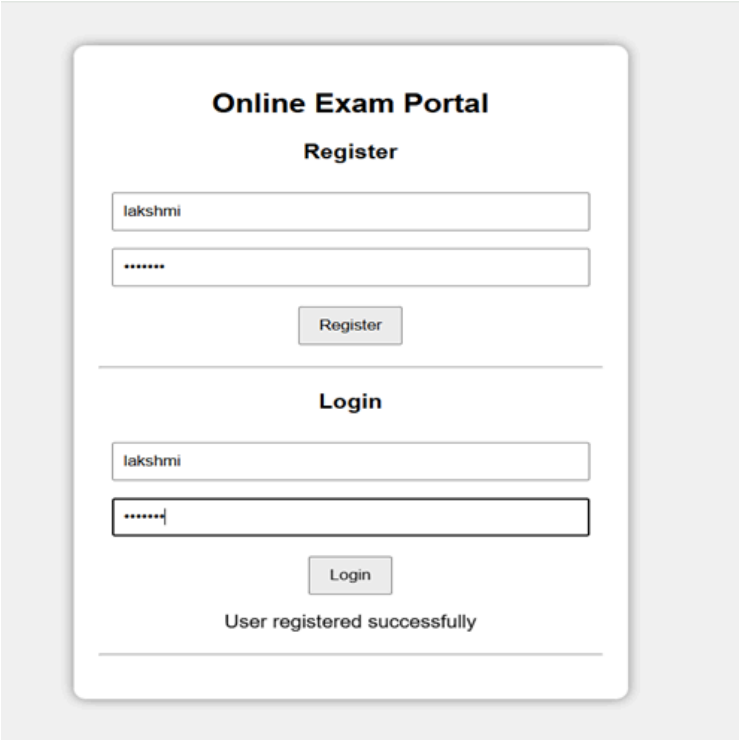
```
        if (selected && selected.value ===
correctAnswers[q]) {
            score++;
        }
    }
    alert("Your Score: " + score + " / " +
total);
function logout() {
    localStorage.removeItem("token");
    location.reload();
}
</script>
</body>
</html>
```

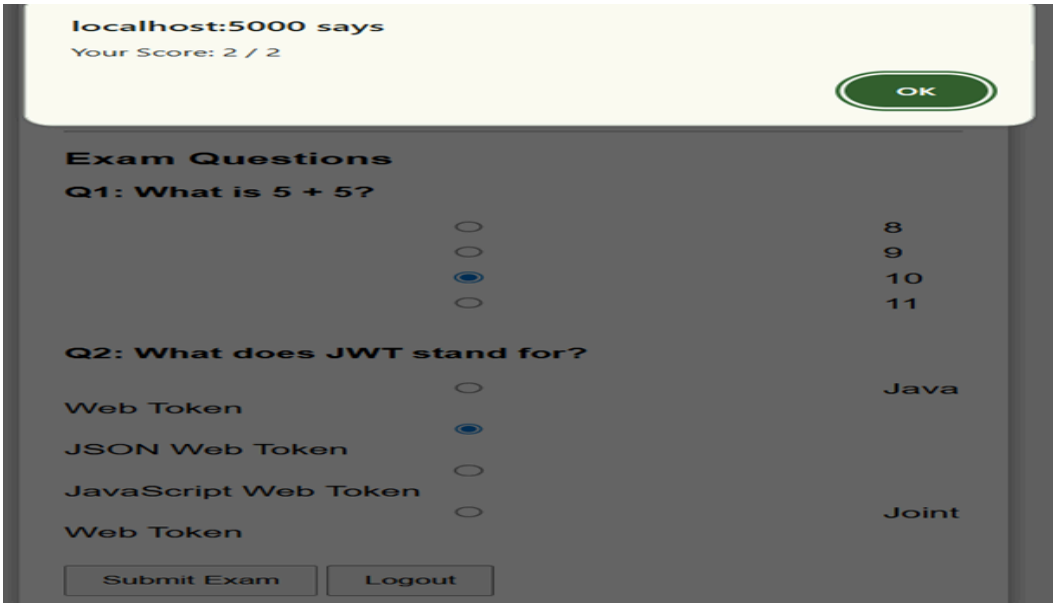
server.js:

```
const express = require("express");
const jwt = require("jsonwebtoken");
const bcrypt = require("bcryptjs");
const cors = require("cors");
const path = require("path");
const app = express();
const PORT = 5000;
const SECRET_KEY = "exam_secret_key";
// Middleware
app.use(express.json());
app.use(cors());
app.use(express.static(__dirname)); // Serves
index.html
// Temporary in-memory storage
let users = [];
app.get("/", (req, res) => {
  res.send("Online Exam Portal API is
Running 🚀");
});
app.post("/register", async (req, res) => {
  const { username, password } = req.body;
  if (!username || !password) {
    return res.status(400).json({ message:
"All fields required" });
  }
  const userExists = users.find(u =>
u.username === username);
  if (userExists) {
    return res.status(400).json({ message:
"User already exists" });
  }
  const hashedPassword = await
bcrypt.hash(password, 10);
  users.push({ username, password:
hashedPassword });
  res.json({ message: "User registered
successfully" });
});
app.post("/login", async (req, res) => {
  const { username, password } = req.body;

  const user = users.find(u => u.username
=== username);
  if (!user) {
    return res.status(400).json({ message:
"Invalid username" });
  }
  const validPassword = await
bcrypt.compare(password, user.password);
  if (!validPassword) {
    return res.status(400).json({ message:
"Invalid password" });
  }
  const token = jwt.sign(
    { username: user.username },
    SECRET_KEY,
    { expiresIn: "1h" }
  );
  res.json({ message: "Login successful",
token });
});
function verifyToken(req, res, next) {
  const header =
req.headers["authorization"];
  const token = header && header.split("
")[1];
  if (!token) {
    return res.status(401).json({ message:
"Access denied. No token provided." });
  }
  jwt.verify(token, SECRET_KEY, (err,
decoded) => {
    if (err) {
      return res.status(403).json({ message:
"Invalid token" });
    }
    req.user = decoded;
    next();
```

```
  });
}
app.get("/exam", verifyToken, (req, res) =>
{
  res.json({
    message: `Welcome
${req.user.username}`,
    exam: [
      {
        id: 1,
        question: "What is 5 + 5?",
        options: ["8", "9", "10", "11"],
        answer: "10"
      },
      {
        id: 2,
        question: "What does JWT stand for?",
        options: [
          "Java Web Token",
          "JSON Web Token",
          "JavaScript Web Token",
          "Joint Web Token"
        ],
        answer: "JSON Web Token"
      }
    ]
  });
});
app.listen(PORT, () => {
  console.log(`Server running at
http://localhost:${PORT}`);
});
```

Output:


Online Exam Portal
Register
lakshmi
Register
Login
lakshmi
Login
User registered successfully


localhost:5000 says
Your Score: 2 / 2
OK
Exam Questions
Q1: What is 5 + 5?
8
9
10
11
Q2: What does JWT stand for?
Java
Web Token
JSON Web Token
JavaScript Web Token
Joint
Web Token
Submit Exam
Logout

12. Placement Registration Website Using React.js Frontend with Node.js Backend APIs

Program Coding:

server.js:

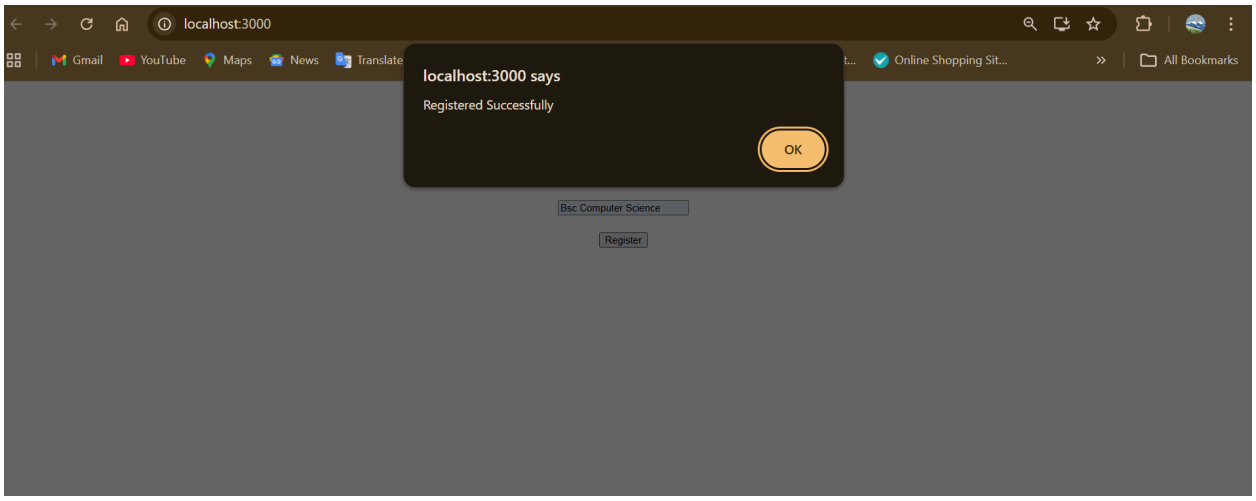
```
const express = require("express");
const cors = require("cors");
const fs = require("fs");
const path = require("path");
const app = express();
app.use(cors());
app.use(express.json());
const filePath = path.join(__dirname,
"students.json");

// Make sure file exists
if (!fs.existsSync(filePath)) {
  fs.writeFileSync(filePath, "[]");
}

app.post("/register", (req, res) => {
  try {
    const newStudent = req.body;

    const data = fs.readFileSync(filePath);
    const students = JSON.parse(data);

    students.push(newStudent);

    fs.writeFileSync(filePath,
JSON.stringify(students, null, 2));

    res.status(200).send("Student Registered
Successfully");
  } catch (error) {
    console.log("ERROR:", error);
    res.status(500).send("Server Error");
  }
});

app.listen(5000, () => {
  console.log("Server running on port
5000");
});
```

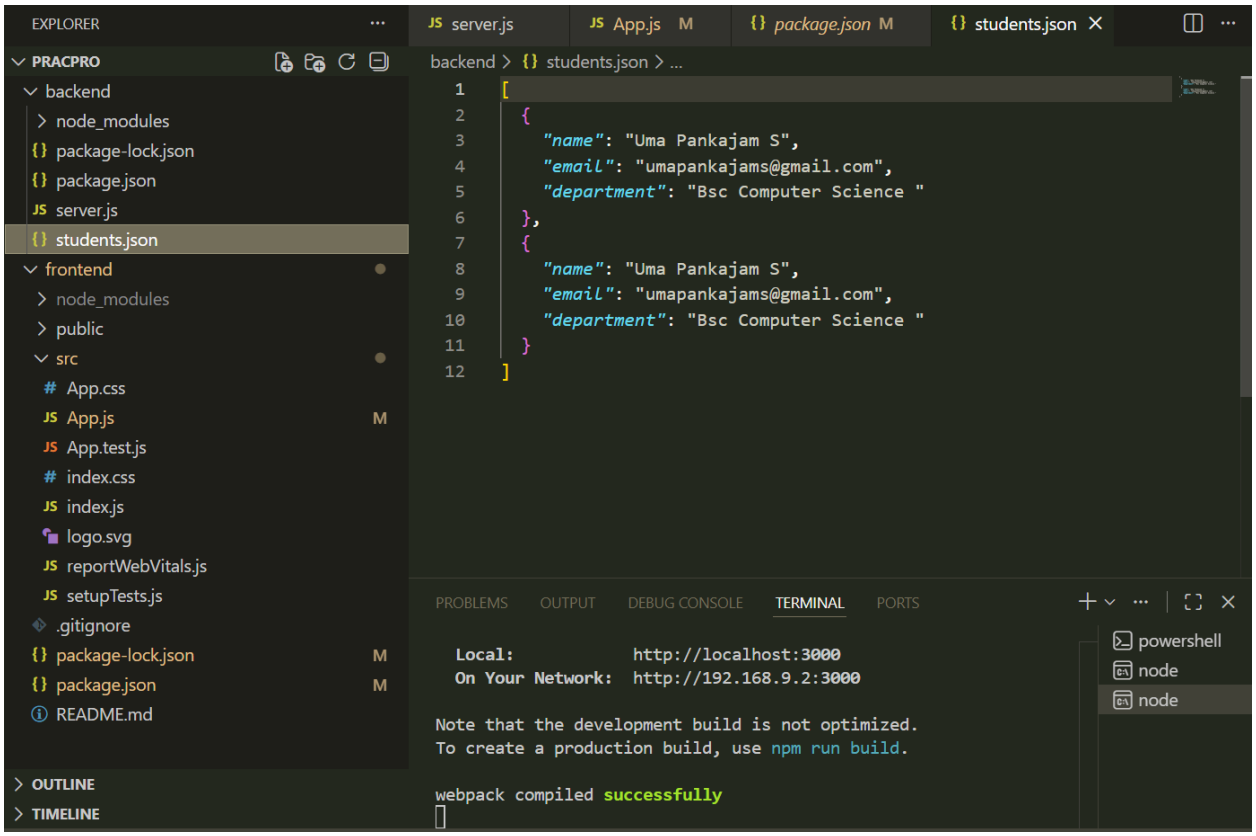
app.js:

```
import React, { useState } from "react";
import axios from "axios";

function App() {
  const [formData, setFormData] =
useState({
    name: "",
    email: "",
    department: ""
  });

  const handleChange = (e) => {
    setFormData({ ...formData,
[e.target.name]: e.target.value });
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    await
axios.post("http://localhost:5000/register",
formData);
    alert("Registered Successfully");
  };

  return (
    <div style={{ textAlign: "center" }}>
      <h2>Placement Registration</h2>
      <form onSubmit={handleSubmit}>
```

```
        <input name="name"
placeholder="Name"
onChange={handleChange} required /><br
/><br />
        <input name="email"
placeholder="Email"
onChange={handleChange} required /><br
/><br />
        <input name="department"
placeholder="Department"
onChange={handleChange} required /><br
/><br />
        <button
type="submit">Register</button>
      </form>
    </div>
  );
}

export default App;
```

Output:


localhost:3000
Gmail
YouTube
Maps
News
Translate
Online Shopping Sit...
All Bookmarks
localhost:3000 says
Registered Successfully
OK
Bsc Computer Science
Register


EXPLORER
PRACPRO
backend
node_modules
package-lock.json
package.json
server.js
students.json
frontend
node_modules
public
src
App.css
App.js
App.test.js
index.css
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
README.md
OUTLINE
TIMELINE
server.js
App.js M
package.json M
students.json
backend > students.json > ...
[
{
"name": "Uma Pankajam S",
"email": "umapankajams@gmail.com",
"department": "Bsc Computer Science "
},
{
"name": "Uma Pankajam S",
"email": "umapankajams@gmail.com",
"department": "Bsc Computer Science "
}
]
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
node
node
Local: http://localhost:3000
On Your Network: http://192.168.9.2:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully

13. Deployment of a Full-Stack Student Admission Application on a Cloud Platform

Program Coding:

app.jsx

```
import { useState, useEffect } from "react";
import axios from "axios";
export default function App() {
  const [name, setName] = useState("");
  const [phone, setPhone] = useState("");
  const [students, setStudents] = useState([]);
  const API = "https://student-admission-backend.onrender.com";
  const fetchStudents = async () => {
    const res = await axios.get(`${API}/users`);
    setStudents(res.data);
  };
  const addStudent = async () => {
    if (!name || !phone) return;
    await axios.post(`${API}/add-user`, { name, phone });
    setName("");
    setPhone("");
    fetchStudents();
  };
  const deleteStudent = async (id) => {
    await axios.delete(`${API}/delete-user/${id}`);
    fetchStudents();
  };
  useEffect(() => {
    fetchStudents();
  }, []);
  return (
    <div style={{ padding: 40 }}>
      <h2>🎓 Student Admission List</h2>
      <input
        placeholder="Student Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <br /><br />
      <input
        placeholder="Phone Number"
        value={phone}
        onChange={(e) => setPhone(e.target.value)}
      />
      <br /><br />
      <button onClick={addStudent}>Add Student</button>
      <h3>Admission Records</h3>
      <ul>
        {students.map((s) => (
          <li key={s.id}>
            {s.name} — {s.phone}
            <button
              style={{ marginLeft: 10 }}
              onClick={() => deleteStudent(s.id)} >
              Remove
            </button>
          </li>
        ))}
      </ul>
    </div>
  ); }
```

main.jsx

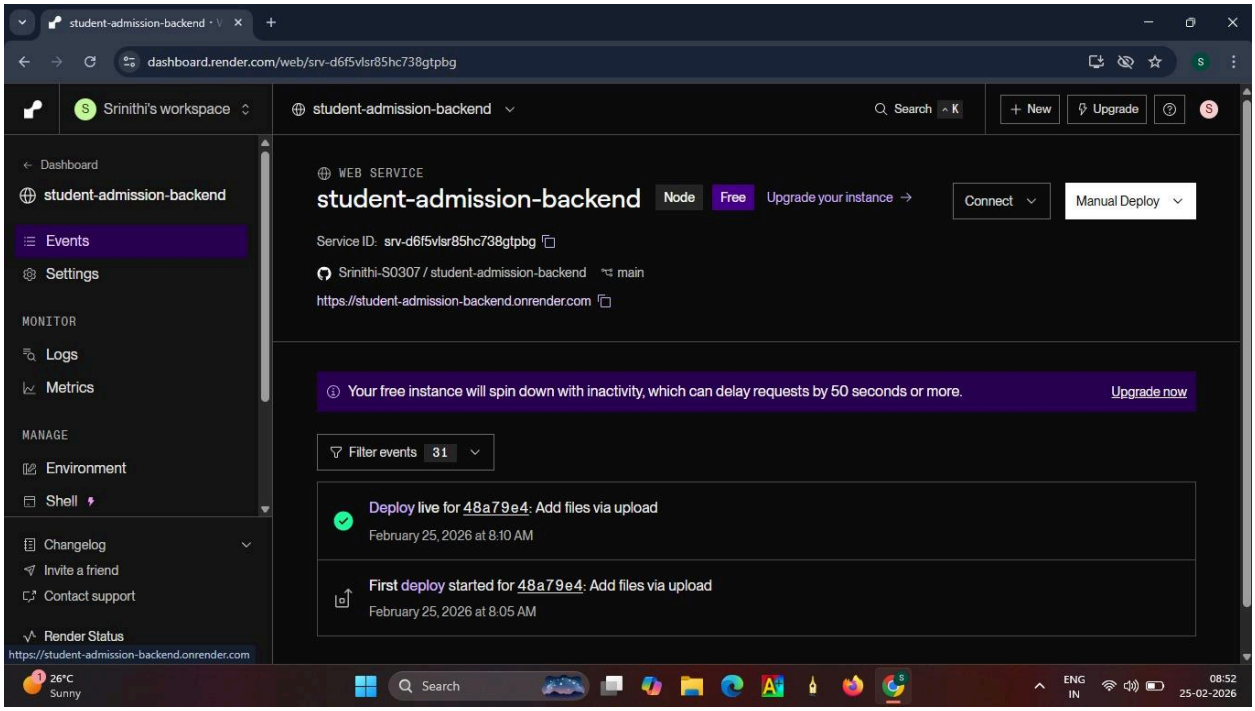
```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.jsx'
```

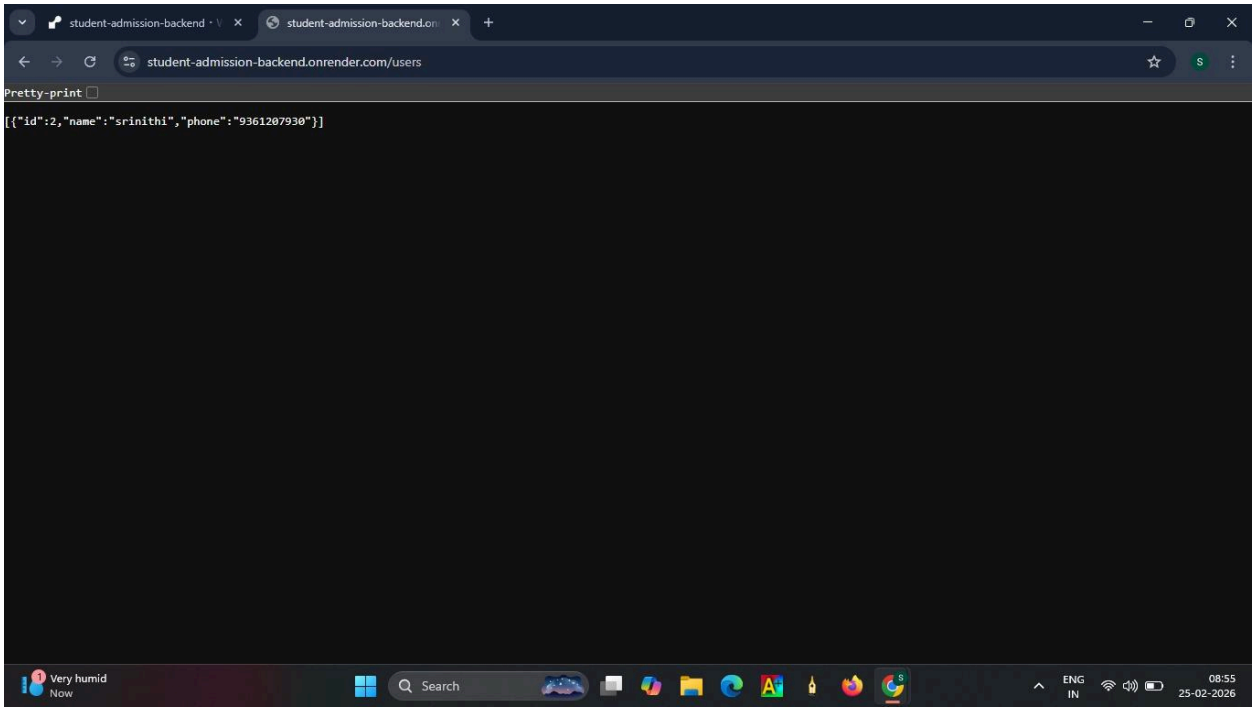
```
createRoot(document.getElementById('root')).render(
  <StrictMode>
    <App />
  </StrictMode>,
)
```

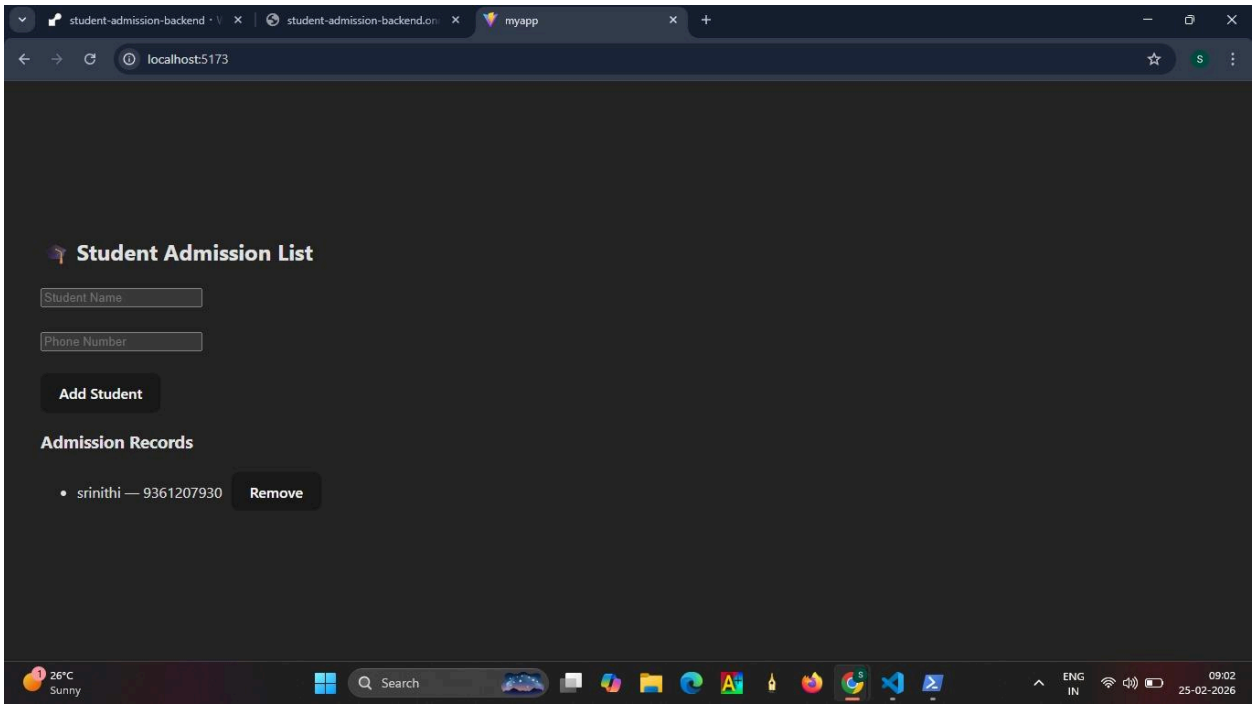
server.js

```
const express = require("express");
const cors = require("cors");
const sqlite3 = require("sqlite3").verbose();
const app = express();
// Middleware
app.use(cors());
app.use(express.json());
// Connect SQLite Database
const db = new sqlite3.Database("./database.db", (err) => {
  if (err) {
    console.log(err.message);
  } else {
    console.log("Connected to SQLite database");
  }
});
// Create Students Table
db.run(`
  CREATE TABLE IF NOT EXISTS students (
      id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT,
    phone TEXT
  )
`);
// ➕ Add Student
app.post("/add-user", (req, res) => {
  const { name, phone } = req.body;
if (!name || !phone) {
    return res.status(400).json({ error: "Name and phone required" });
  }
  db.run(
      "INSERT INTO students(name, phone) VALUES(?, ?)",
    [name, phone],
    function (err) {
      if (err) {
        return res.status(500).json(err);
      }
      res.json({
            message: "Student added successfully",
        id: this.lastID,
      });
    }
  );
});
// 📋 Get All Students
app.get("/users", (req, res) => {
        db.all("SELECT * FROM students", [], (err, rows) => {
    if (err) {
      return res.status(500).json(err);
    }
    res.json(rows);
  });
});
// ❌ Delete Student
app.delete("/delete-user/:id", (req, res) => {
  const id = req.params.id;
    db.run("DELETE FROM students WHERE id = ?", [id], function (err) {
    if (err) {
      return res.status(500).json(err);
    }
        res.json({ message: "Student removed successfully" });
  });
});
// Start Server
app.listen(5000, () => {
    console.log("Backend running on http://localhost:5000");
});
```

Output:


student-admission-backend
dashboard.render.com/web/srv-d6f5vlsr85hc738gtpbg
Srinithi's workspace
student-admission-backend
Search
New
Upgrade
Dashboard
student-admission-backend
Events
Settings
MONITOR
Logs
Metrics
MANAGE
Environment
Shell
Changelog
Invite a friend
Contact support
Render Status
https://student-admission-backend.onrender.com
WEB SERVICE
student-admission-backend
Node
Free
Upgrade your instance
Connect
Manual Deploy
Service ID: srv-d6f5vlsr85hc738gtpbg
Srinithi-S0307 / student-admission-backend
main
https://student-admission-backend.onrender.com
Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.
Upgrade now
Filter events
31
Deploy live for 48a79e4: Add files via upload
February 25, 2026 at 8:10 AM
First deploy started for 48a79e4: Add files via upload
February 25, 2026 at 8:05 AM
26°C
Sunny
Search
ENG
IN
08:52
25-02-2026


student-admission-backend
student-admission-backend.onrender.com/users
Pretty-print
[{"id":2,"name":"srinithi","phone":"9361207930"}]
Very humid
Now
Search
ENG
IN
08:55
25-02-2026


student-admission-backend
student-admission-backend.on
myapp
localhost:5173
Student Admission List
Student Name
Phone Number
Add Student
Admission Records
srinithi — 9361207930
Remove
26°C
Sunny
Search
ENG
IN
09:02
25-02-2026

14. Performance Profiling and Debugging of a Single Page Application Using React.js

Program Coding:

main.jsx:

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./App.css";

ReactDOM.createRoot(document.getElemen
tById("root")).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

App.jsx:

```
import { useState } from "react";
import Counter from "./Counter";
import ExpensiveComponent from
"./ExpensiveComponent";
import "./App.css";

function App() {
  const [count, setCount] = useState(0);
  const [text, setText] = useState("");

  console.log("App Rendered");

  return (
    <div className="container">
      <h1>⚡ Performance Debugging
Demo</h1>

      <Counter count={count}
setCount={setCount} />

      <input
        type="text"
        placeholder="Type something..."
        value={text}
        onChange={(e) =>
setText(e.target.value)}
      />

      <ExpensiveComponent />
    </div>
  );
}

export default App;
```

Counter.jsx

```
function Counter({ count, setCount }) {
  console.log("Counter Rendered");

  return (
    <div className="box">
      <h2>Count: {count}</h2>
      <button onClick={() => setCount(count
+ 1)}>
        Increase
      </button>
    </div>
  );
}

export default Counter;
```

ExpensiveComponent.jsx (Before Optimization)

```
function ExpensiveComponent() {
  console.log("Expensive Component
Rendered");

  let total = 0;
  for (let i = 0; i < 100000000; i++) {
```

```
  total += i;
 }

 return (
  <div className="box">
   <h3>Heavy Calculation Done</h3>
  </div>
 );
}

export default ExpensiveComponent;
```

App,css:

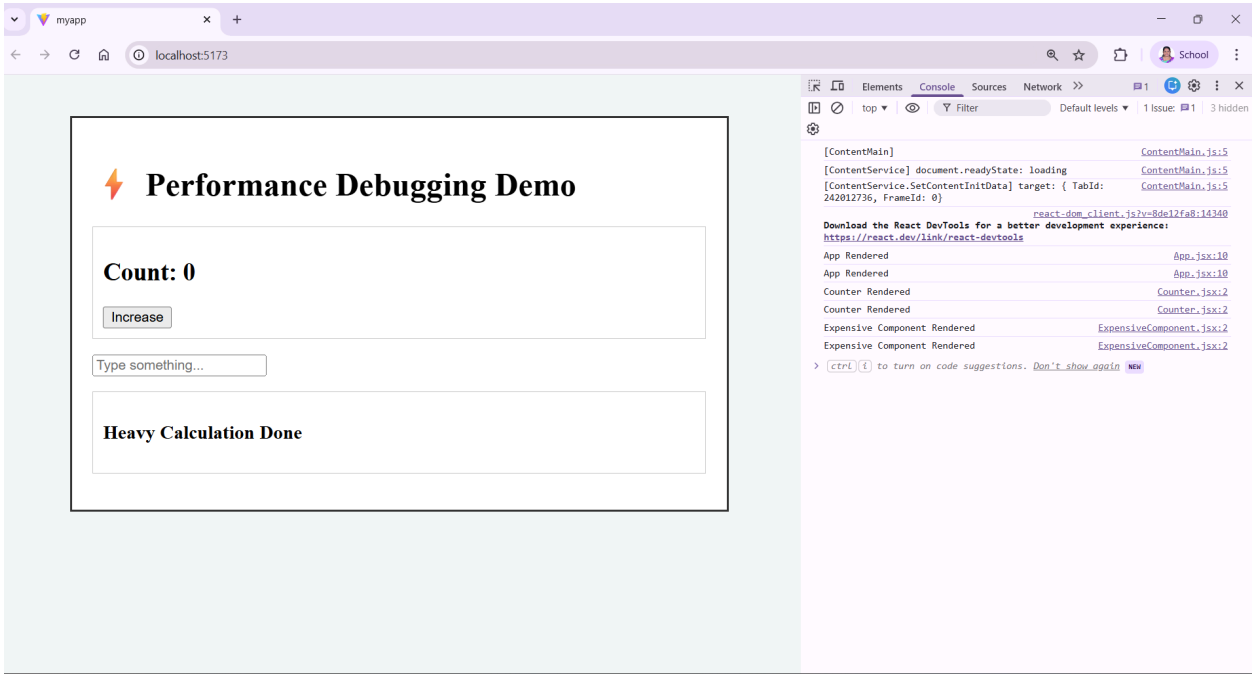
```
body {
 background-color: #f4f6f8;
}

.container {
 width: 600px;
 margin: 40px auto;
 padding: 20px;
 border: 2px solid #333;
 background-color: white;
}

.box {
 margin: 15px 0;
 padding: 10px;
 border: 1px solid #ccc;
}
```

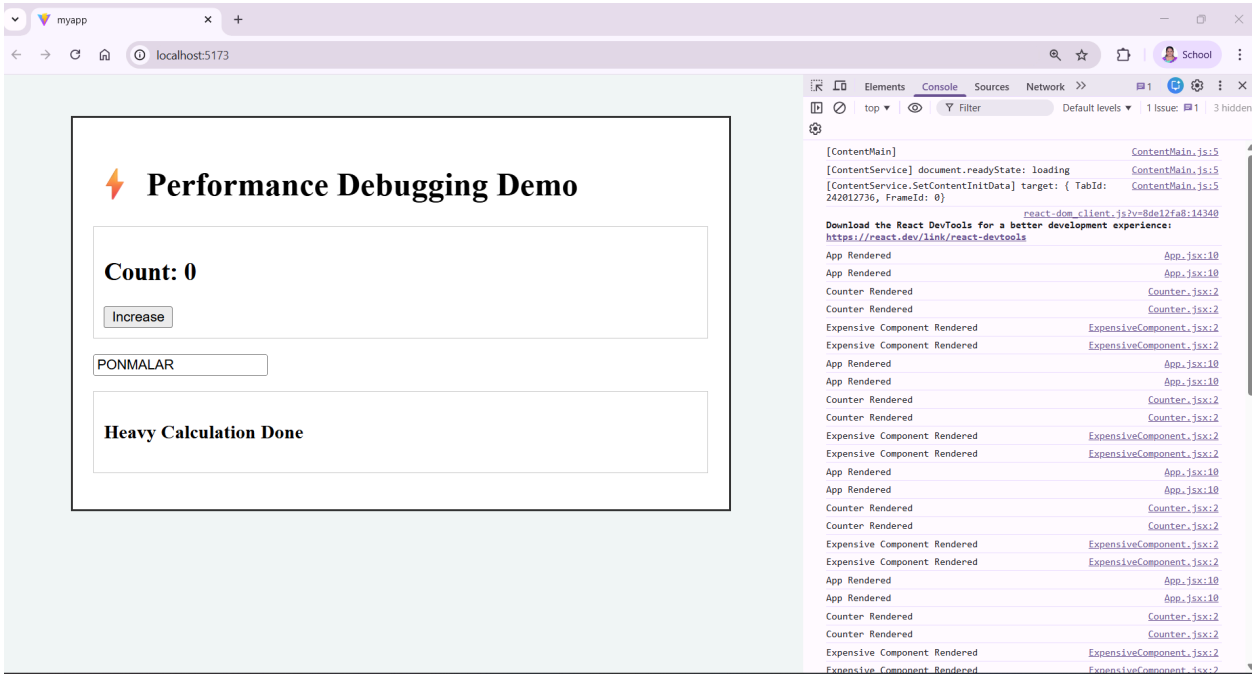
//Optimization Using React.memo

ExpensiveComponent.jsx (After Optimization)

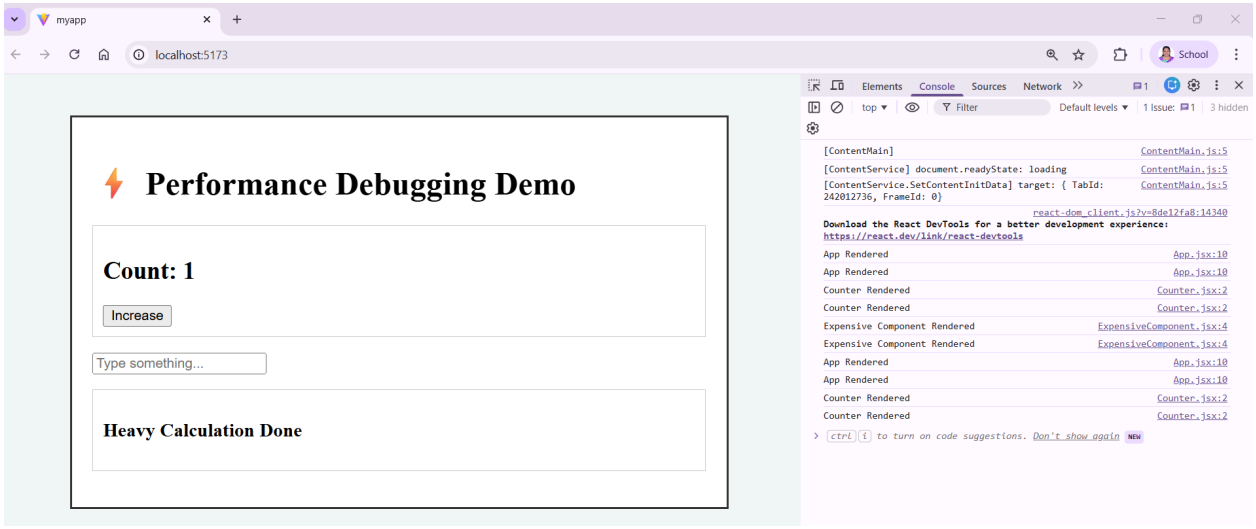
```
import React from "react";

function ExpensiveComponent() {
 console.log("Expensive Component Rendered");

 let total = 0;
 for (let i = 0; i < 100000000; i++) {
  total += i;
 }

 return (
  <div className="box">
   <h3>Heavy Calculation Done</h3>
  </div>
 );
}

export default React.memo(ExpensiveComponent);
```

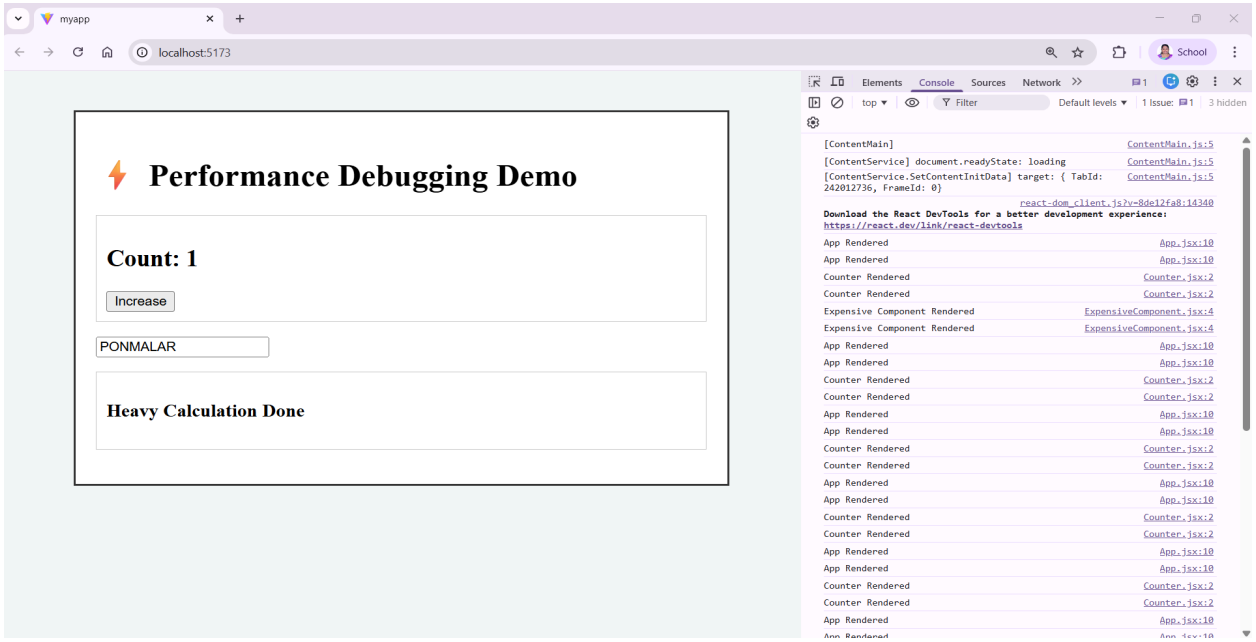
Output:
(Before Optimization)





(After Optimization)





Performance Comparison

Feature	Before	After
Expensive Component Render	Every time	Only once
UI Speed	Slow	Smooth
Heavy Loop Execution	Multiple times	Single time

15. React To-Do App with Git Version Control

Git deployment

PART 1 — Initial Setup (Project Owner)

Step 1: Install Requirements

Make sure you have:

- Node.js
- Git
- VS Code
- GitHub account

Check versions:

```
node -v
git --version
```

Step 2: Create React App

Open VS Code → Open Terminal:

```
npx create-react-app react-todo-app
cd react-todo-app
code .
```

Start app:

```
npm start
```

Step 3: Initialize Git

Inside project folder:

```
git init
```

Check status:

```
git status
```

Step 4: First Commit

```
git add .
git commit -m "Initial React app setup"
```

Step 5: Create GitHub Repository

Go to GitHub → New Repository →

Name it:

```
react-todo-app
```

Then connect local project:

```
git remote add origin
https://github.com/yourusername/react-todo-app.git
git branch -M main
git push -u origin main
```

Program Coding:

App.jsx:

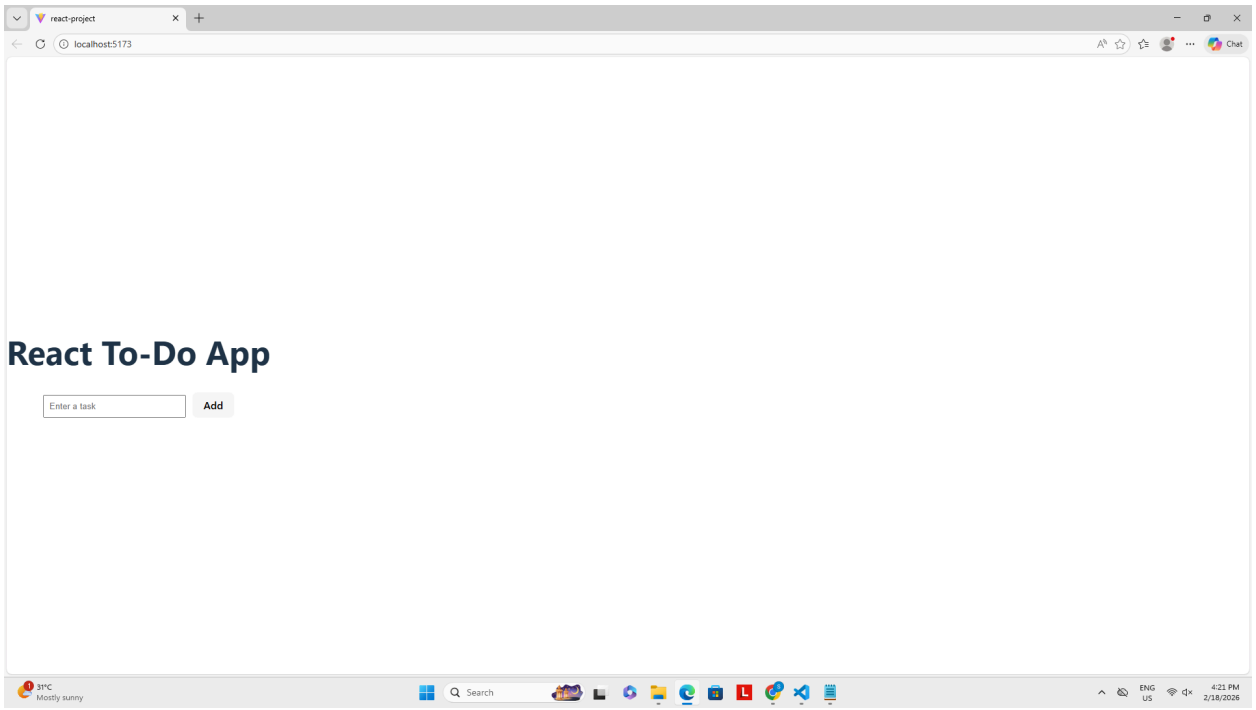
```
  import { useState } from "react";

function App() {
 // State for current input
 const [task, setTask] = useState("");
 // State for list of tasks
 const [tasks, setTasks] = useState([]);

 // Add task
 const addTask = () => {
  if (task.trim() === "") return;
  setTasks([...tasks, task]);
  setTask("");
 };

 // Delete task
 const deleteTask = (indexToDelete) => {
  setTasks(tasks.filter((_, index) => index
!== indexToDelete));
 };

 return (
  <div style={{ textAlign: "center",
marginTop: "50px" }}>
   <h1>React To-Do App</h1>

   <div style={{ marginBottom: "20px"
}}>
    <input
     type="text"
     placeholder="Enter a task"
     value={task}
     onChange={(e) =>
setTask(e.target.value)}
     style={{ padding: "8px", width:
"200px" }}
    />
    <button
     onClick={addTask}
     style={{
      padding: "8px 16px",
      marginLeft: "10px",
```

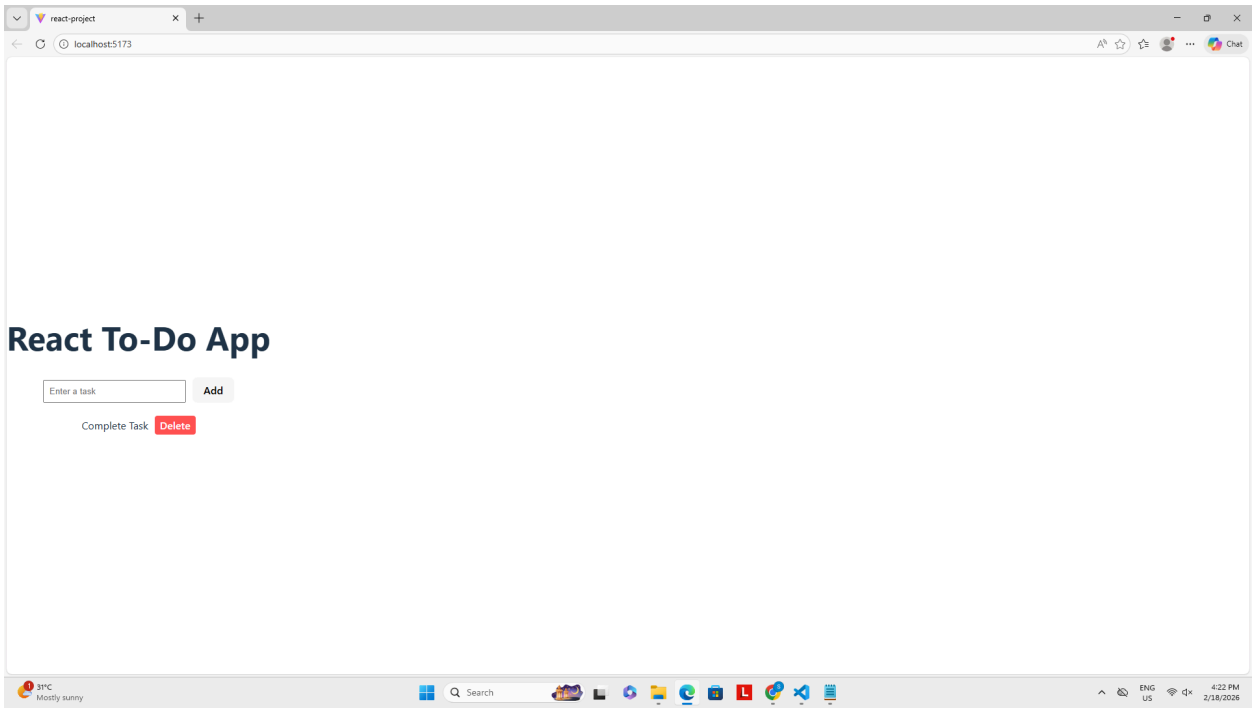
```
          cursor: "pointer",
        }}
      >
        Add
      </button>
    </div>

    <ul style={{ listStyleType: "none",
padding: 0 }}>
      {tasks.map((t, index) => (
        <li
          key={index}
          style={{
            marginBottom: "10px",
            display: "flex",
            justifyContent: "center",
            alignItems: "center",
            gap: "10px",
          }}
        >
          <span>{t}</span>
          <button
            onClick={() => deleteTask(index)}
            style={{
              padding: "4px 8px",
              cursor: "pointer",
              backgroundColor: "#ff4d4f",
              color: "white",
              border: "none",
              borderRadius: "4px",
            }}
          >
            Delete
          </button>
        </li>
      ))}
    </ul>
  </div>
 );
}

export default A
```

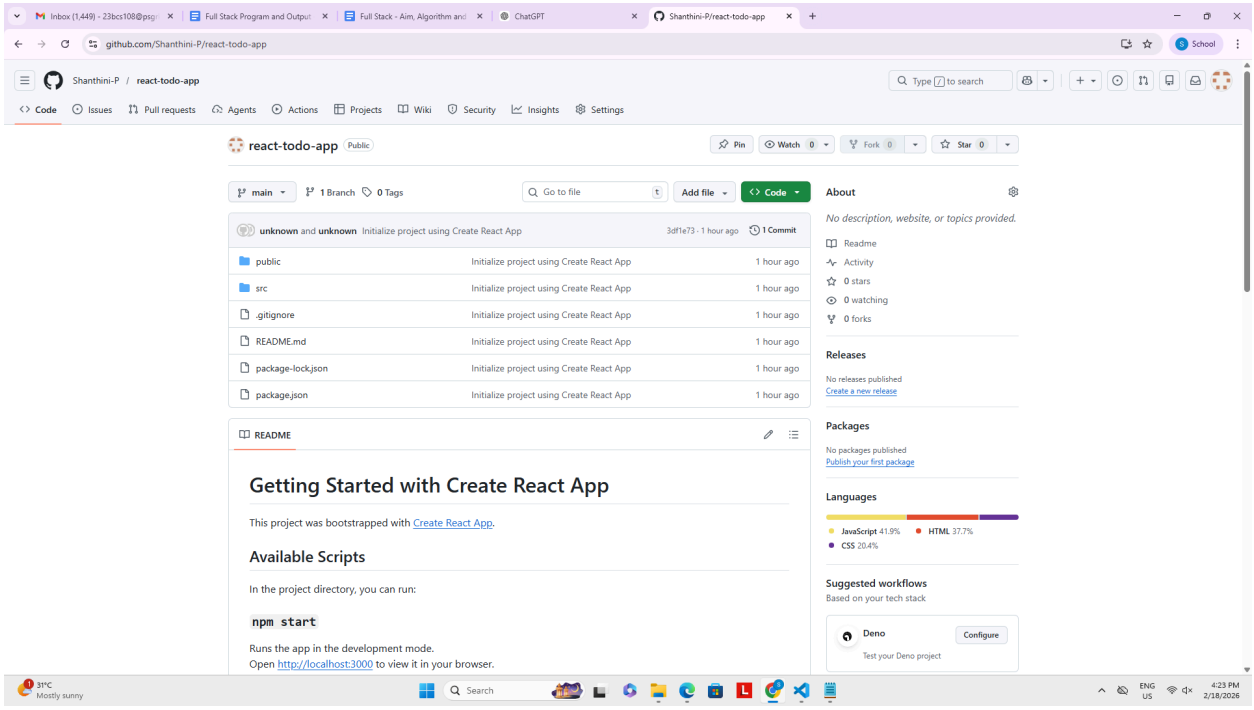
Main.jsx:

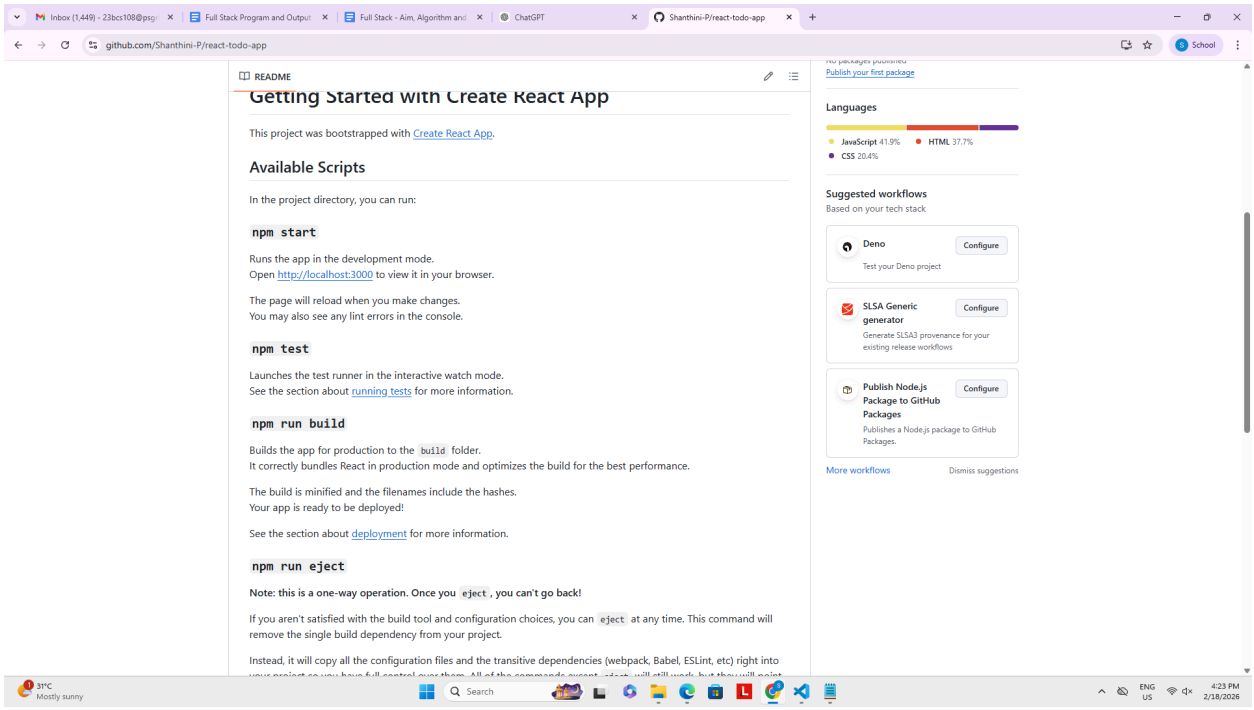
```
    import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./index.css"; // optional styling

ReactDOM.createRoot(document.getElemen
tById("root")).render(
 <React.StrictMode>
  <App />
 </React.StrictMode>
);
```

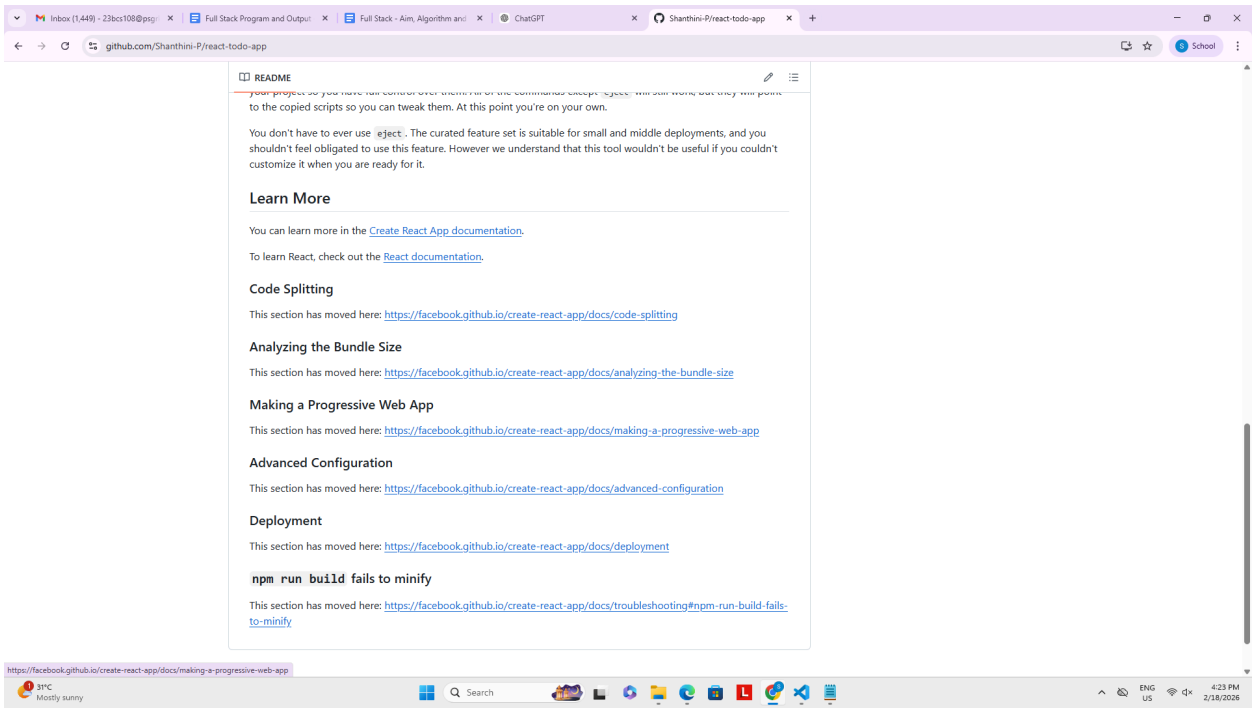
Output:


react-project
localhost:5173
Chat
React To-Do App
Enter a task
Add
31°C
Mostly sunny
Search
ENG
US
4:21 PM
2/18/2026


react-project
localhost:5173
Chat
React To-Do App
Enter a task
Add
Complete Task
Delete
31°C
Mostly sunny
Search
ENG
US
4:22 PM
2/18/2026


react-todo-app
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
unknown and unknown Initialize project using Create React App
3df1e73 · 1 hour ago
1 Commit
public
src
.gitignore
README.md
package-lock.json
package.json
Initialize project using Create React App
1 hour ago
README
Getting Started with Create React App
This project was bootstrapped with Create React App.
Available Scripts
In the project directory, you can run:
npm start
Runs the app in the development mode.
Open http://localhost:3000 to view it in your browser.
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Languages
JavaScript 41.9%
HTML 37.7%
CSS 20.4%
Suggested workflows
Based on your tech stack
Deno
Configure
Test your Deno project


README
Getting Started with Create React App
This project was bootstrapped with Create React App.
Available Scripts
In the project directory, you can run:
npm start
Runs the app in the development mode.
Open http://localhost:3000 to view it in your browser.
The page will reload when you make changes.
You may also see any lint errors in the console.
npm test
Launches the test runner in the interactive watch mode.
See the section about running tests for more information.
npm run build
Builds the app for production to the build folder.
It correctly bundles React in production mode and optimizes the build for the best performance.
The build is minified and the filenames include the hashes.
Your app is ready to be deployed!
See the section about deployment for more information.
npm run eject
Note: this is a one-way operation. Once you eject, you can't go back!
If you aren't satisfied with the build tool and configuration choices, you can eject at any time. This command will remove the single build dependency from your project.
Instead, it will copy all the configuration files and the transitive dependencies (webpack, Babel, ESLint, etc) right into
Publish your first package
Languages
JavaScript 41.9%
HTML 37.7%
CSS 20.4%
Suggested workflows
Based on your tech stack
Deno
Configure
Test your Deno project
SLSA Generic generator
Configure
Generate SLSA3 provenance for your existing release workflows
Publish Node.js Package to GitHub Packages
Configure
Publishes a Node.js package to GitHub Packages.
More workflows
Dismiss suggestions


Inbox (1,449) - 23bcs108@psg
Full Stack Program and Output
Full Stack - Aim, Algorithm and
ChatGPT
Shanthini-P/react-todo-app
github.com/Shanthini-P/react-todo-app
School
README
to the copied scripts so you can tweak them. At this point you're on your own.
You don't have to ever use eject. The curated feature set is suitable for small and middle deployments, and you shouldn't feel obligated to use this feature. However we understand that this tool wouldn't be useful if you couldn't customize it when you are ready for it.
Learn More
You can learn more in the Create React App documentation.
To learn React, check out the React documentation.
Code Splitting
This section has moved here: https://facebook.github.io/create-react-app/docs/code-splitting
Analyzing the Bundle Size
This section has moved here: https://facebook.github.io/create-react-app/docs/analyzing-the-bundle-size
Making a Progressive Web App
This section has moved here: https://facebook.github.io/create-react-app/docs/making-a-progressive-web-app
Advanced Configuration
This section has moved here: https://facebook.github.io/create-react-app/docs/advanced-configuration
Deployment
This section has moved here: https://facebook.github.io/create-react-app/docs/deployment
npm run build fails to minify
This section has moved here: https://facebook.github.io/create-react-app/docs/troubleshooting#npm-run-build-fails-to-minify
https://facebook.github.io/create-react-app/docs/making-a-progressive-web-app
31°C
Mostly sunny
Search
ENG
US
4:23 PM
2/18/2026

16. Build and Deploy a Full Stack Web Application – Online Shopping

Program Coding:

Backend:

```
mkdir backend
cd backend
npm init -y

npm install express mongoose cors
npm install nodemon --save-dev
```

server.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());

mongoose.connect("YOUR_MONGODB_ATLAS_URL")
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));

// Product Schema
const Product = mongoose.model("Product", {
  name: String,
  price: Number,
  image: String
});

// Get products
app.get("/products", async (req, res) => {
  const products = await Product.find();
  res.json(products);
});

// Add product (for testing)
app.post("/products", async (req, res) => {
  const newProduct = new Product(req.body);
  await newProduct.save();
  res.json(newProduct);
});

app.listen(5000, () => console.log("Server running on port 5000"));
```

Run backend:

```
npx nodemon server.js
```

Frontend:

```
npx create-react-app frontend
cd frontend
npm install axios
```

App.js:

```
import React, { useState, useEffect } from "react";
import "./App.css";

const productsData = [
  { id: 1, name: "Shoes", price: 1500, image: "https://via.placeholder.com/150" },
  { id: 2, name: "T-Shirt", price: 800, image: "https://via.placeholder.com/150" },
  { id: 3, name: "Watch", price: 2500, image: "https://via.placeholder.com/150" },
];

function App() {
```

```
  const [cart, setCart] = useState([]);
  const [search, setSearch] = useState("");
  const [showCheckout, setShowCheckout]
= useState(false);

  // Load cart from localStorage
  useEffect(() => {
    const savedCart =
JSON.parse(localStorage.getItem("cart"));
    if (savedCart) setCart(savedCart);
  }, []);

  // Save cart to localStorage
  useEffect(() => {
    localStorage.setItem("cart",
JSON.stringify(cart));
  }, [cart]);

  const addToCart = (product) => {
    const existing = cart.find((item) =>
item.id === product.id);
    if (existing) {
      setCart(cart.map((item) =>
        item.id === product.id
          ? { ...item, quantity: item.quantity + 1
}
          : item
      ));
    } else {
      setCart([...cart, { ...product, quantity: 1
}]);
    }
  };

  const increaseQty = (id) => {
    setCart(cart.map(item =>
      item.id === id ? { ...item, quantity:
item.quantity + 1 } : item
    ));
  };

  const decreaseQty = (id) => {
    setCart(cart.map(item =>
      item.id === id && item.quantity > 1
        ? { ...item, quantity: item.quantity - 1 }
        : item
    ));
  };

  const removeItem = (id) => {
    setCart(cart.filter(item => item.id !==
id));
  };

  const totalAmount = cart.reduce(
    (total, item) => total + item.price *
item.quantity,
    0
  );

  const filteredProducts =
productsData.filter(product =>

product.name.toLowerCase().includes(searc
h.toLowerCase())
  );

  return (
    <div className="container">
      <h1>🛒 Online Shopping</h1>

      <input
        type="text"
        placeholder="Search product..."
        value={search}
        onChange={(e) =>
setSearch(e.target.value)}
        className="search"
      />

      <div className="products">
        {filteredProducts.map(product => (
```

```
        <div key={product.id}
className="card">
          <img src={product.image}
alt={product.name} />
          <h3>{product.name}</h3>
          <p>₹{product.price}</p>
          <button onClick={() =>
addToCart(product)}>Add to Cart</button>
        </div>
       ))}
     </div>

     <h2>Cart</h2>

     {cart.length === 0 ? (
       <p>Cart is Empty</p>
     ) : (
       <>
         {cart.map(item => (
           <div key={item.id}
className="cart-item">
             <span>{item.name}</span>
             <div>
               <button onClick={() =>
decreaseQty(item.id)}>-</button>
               <span>{item.quantity}</span>
               <button onClick={() =>
increaseQty(item.id)}>+</button>
             </div>
             <span>₹{item.price *
item.quantity}</span>
             <button onClick={() =>
removeItem(item.id)}>Remove</button>
           </div>
         ))}

         <h3>Total: ₹{totalAmount}</h3>

         <button className="checkout-btn"
onClick={() => setShowCheckout(true)}>
           Proceed to Checkout
         </button>
       </>
     )}

     {showCheckout && (
       <div className="checkout">
         <h2>Checkout</h2>
         <input type="text" placeholder="Full
Name" />
         <input type="text"
placeholder="Address" />
         <button onClick={() => alert("Order
Placed Successfully!")}>
           Place Order
         </button>
       </div>
     )}
   </div>
  );
}

export default App;
```

App.css:

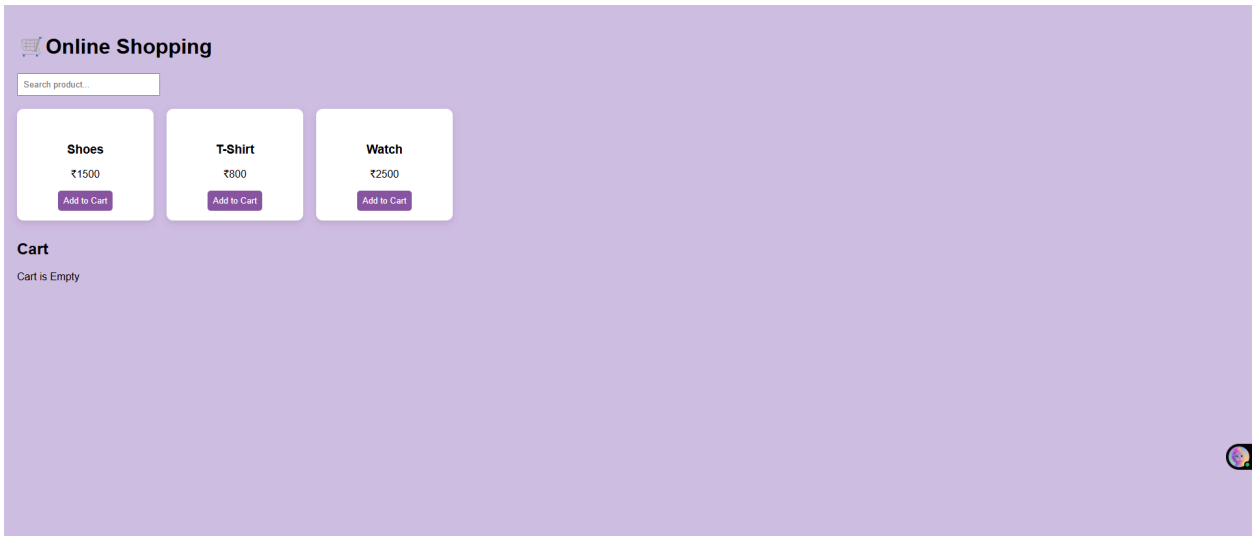
```
body {
  font-family: Arial, sans-serif;
  background-color: #d0bee4;
}

.container {
  padding: 20px;
}

.search {
  padding: 8px;
  margin-bottom: 20px;
  width: 200px;
}

.products {
  display: flex;
```

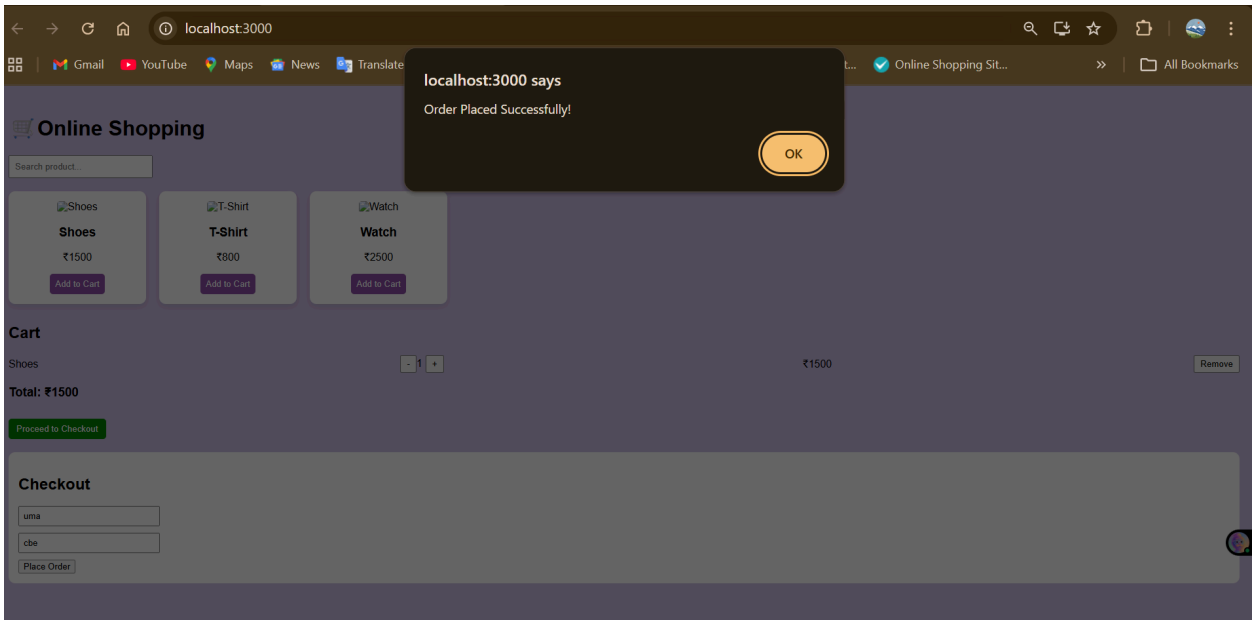
```
  gap: 20px;
  margin-bottom: 30px;
}

.card {
  background: white;
  padding: 15px;
  border-radius: 10px;
  box-shadow: 0 4px 10px rgba(0,0,0,0.1);
  width: 180px;
  text-align: center;
}

.card img {
  width: 100%;
  border-radius: 5px;
}

.card button {
  background-color: #8b55a4;
  color: white;
  border: none;
  padding: 8px;
  border-radius: 5px;
  cursor: pointer;
}

.card button:hover {
  background-color: #0056b3;
}

.cart-item {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 10px;
}

.cart-item button {
  margin-left: 5px;
  padding: 4px 8px;
}

.checkout-btn {
  margin-top: 10px;
  background-color: green;
  color: white;
  padding: 8px 12px;
  border: none;
  border-radius: 5px;
}

.checkout {
  margin-top: 20px;
  background: white;
  padding: 15px;
  border-radius: 10px;
}

.checkout input {
  display: block;
  margin-bottom: 10px;
  padding: 6px;
  width: 200px;
}
```

Run Frontend:

```
npm start
```

Output:


Online Shopping
Search product...
Shoes
₹1500
Add to Cart
T-Shirt
₹800
Add to Cart
Watch
₹2500
Add to Cart
Cart
Cart is Empty


localhost:3000
Gmail
YouTube
Maps
News
Translate
Online Shopping Sit...
All Bookmarks
localhost:3000 says
Order Placed Successfully!
OK
Online Shopping
Search product...
Shoes
₹1500
Add to Cart
T-Shirt
₹800
Add to Cart
Watch
₹2500
Add to Cart
Cart
Shoes
1
₹1500
Remove
Total: ₹1500
Proceed to Checkout
Checkout
uma
cbe
Place Order